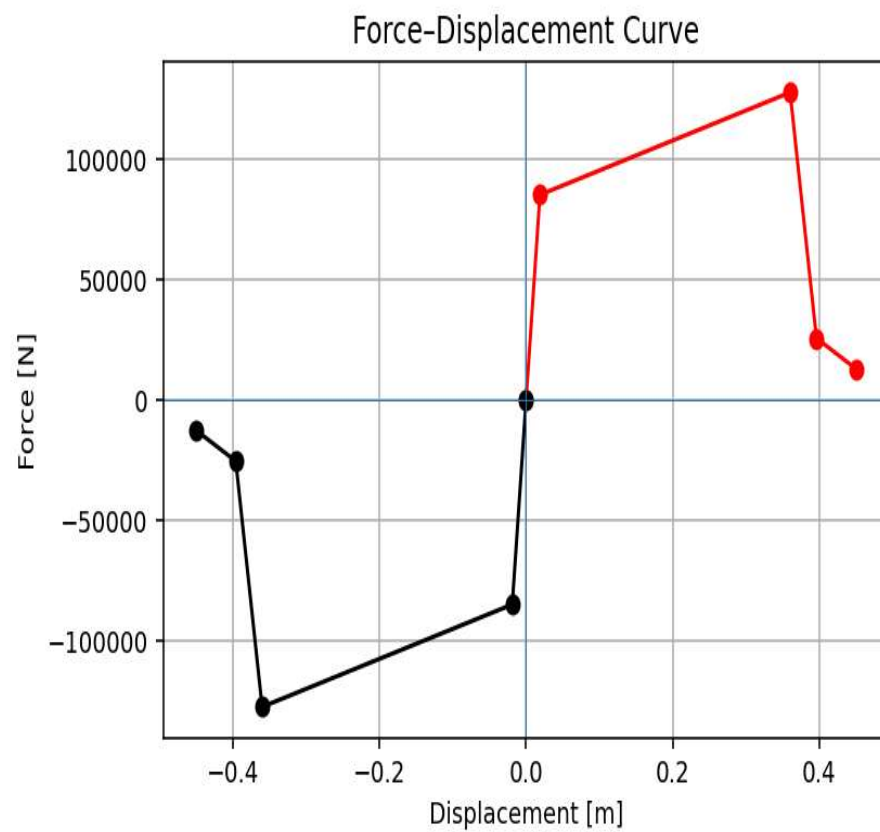
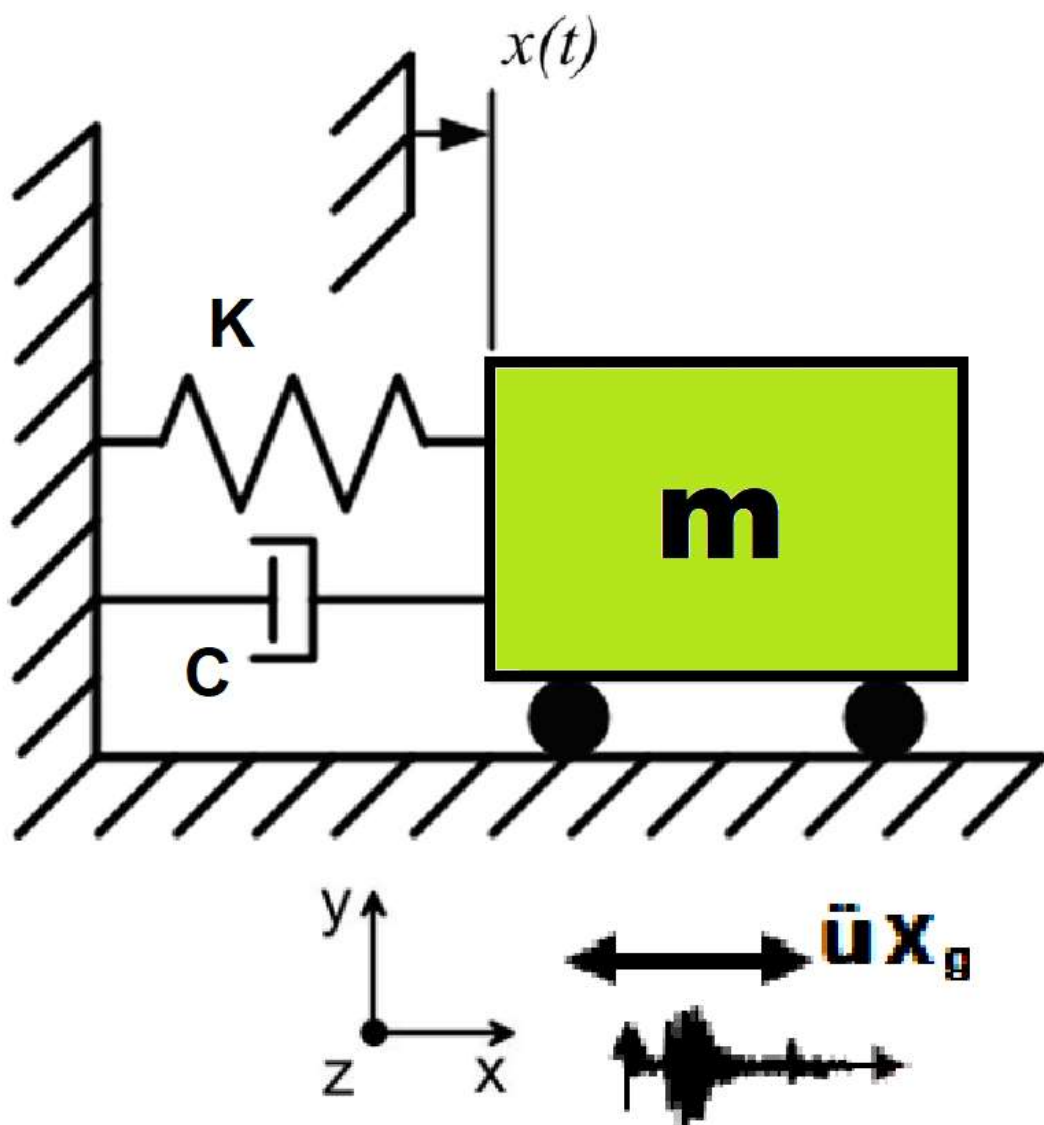


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

NONLINEAR DYNAMIC ANALYSIS UNDER A SINGLE GROUND MOTION RECORD COMPUTATION AND VISUALIZATION RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OPENSEES

(CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM)

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

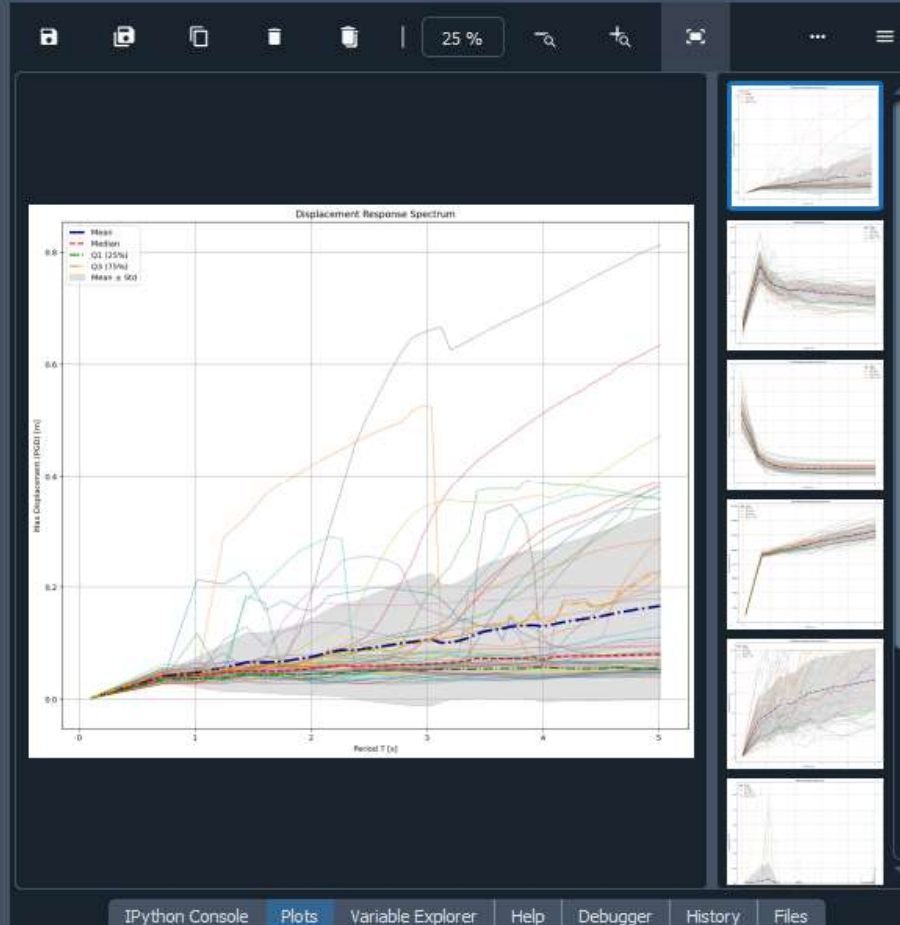
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

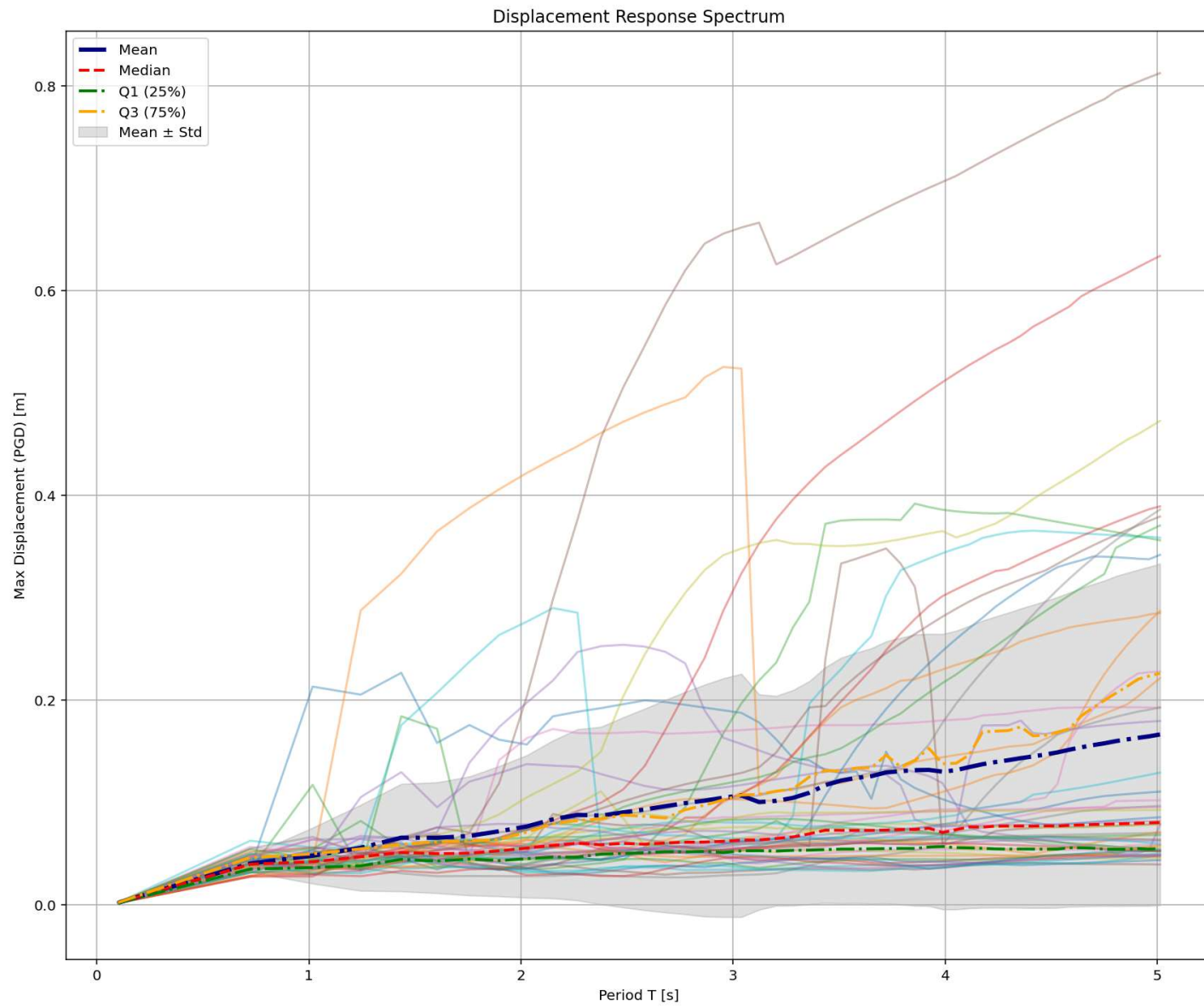
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SDOF_RESPONSE_SPEC...EISMIC_DUCT_OSF.py X
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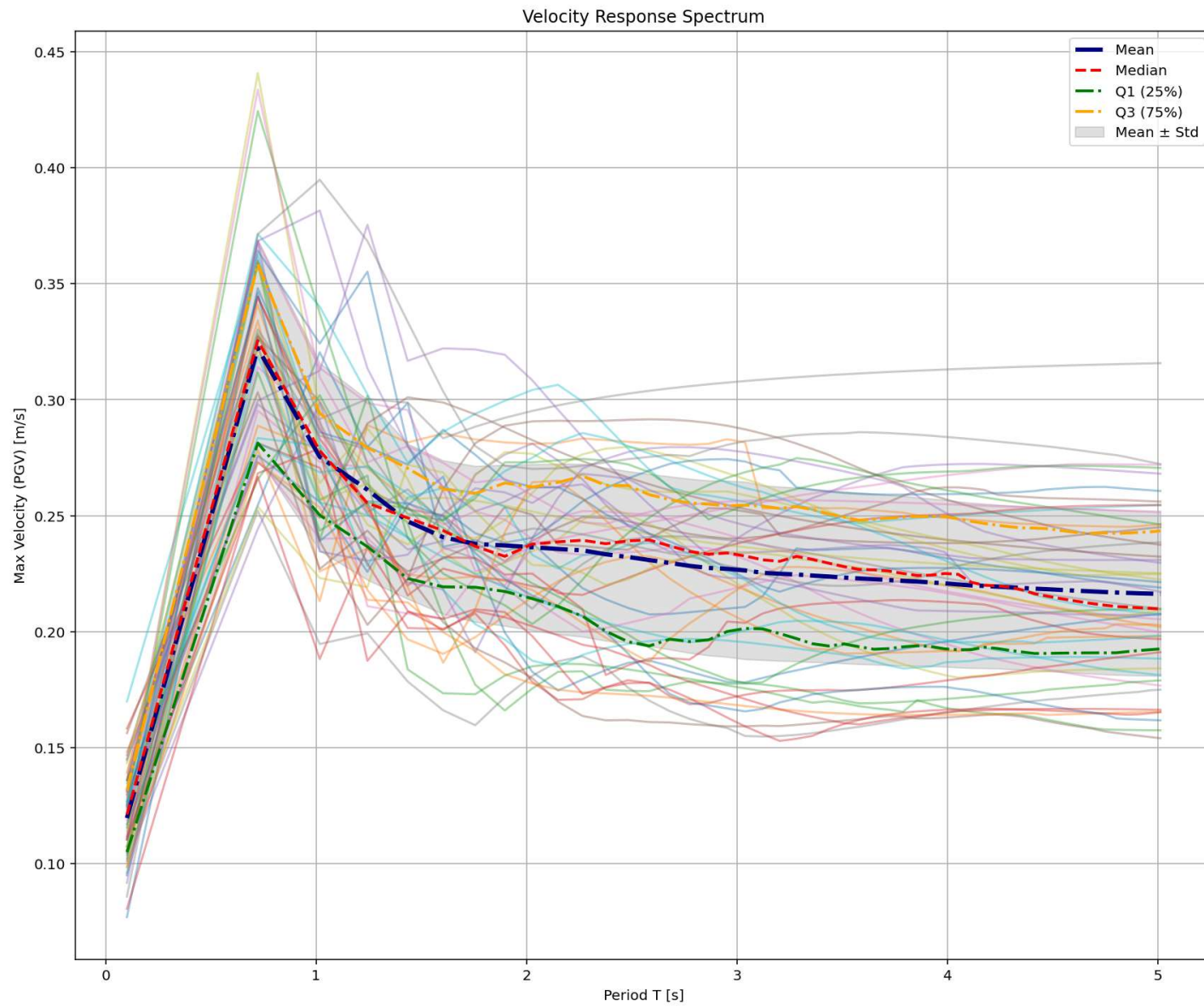
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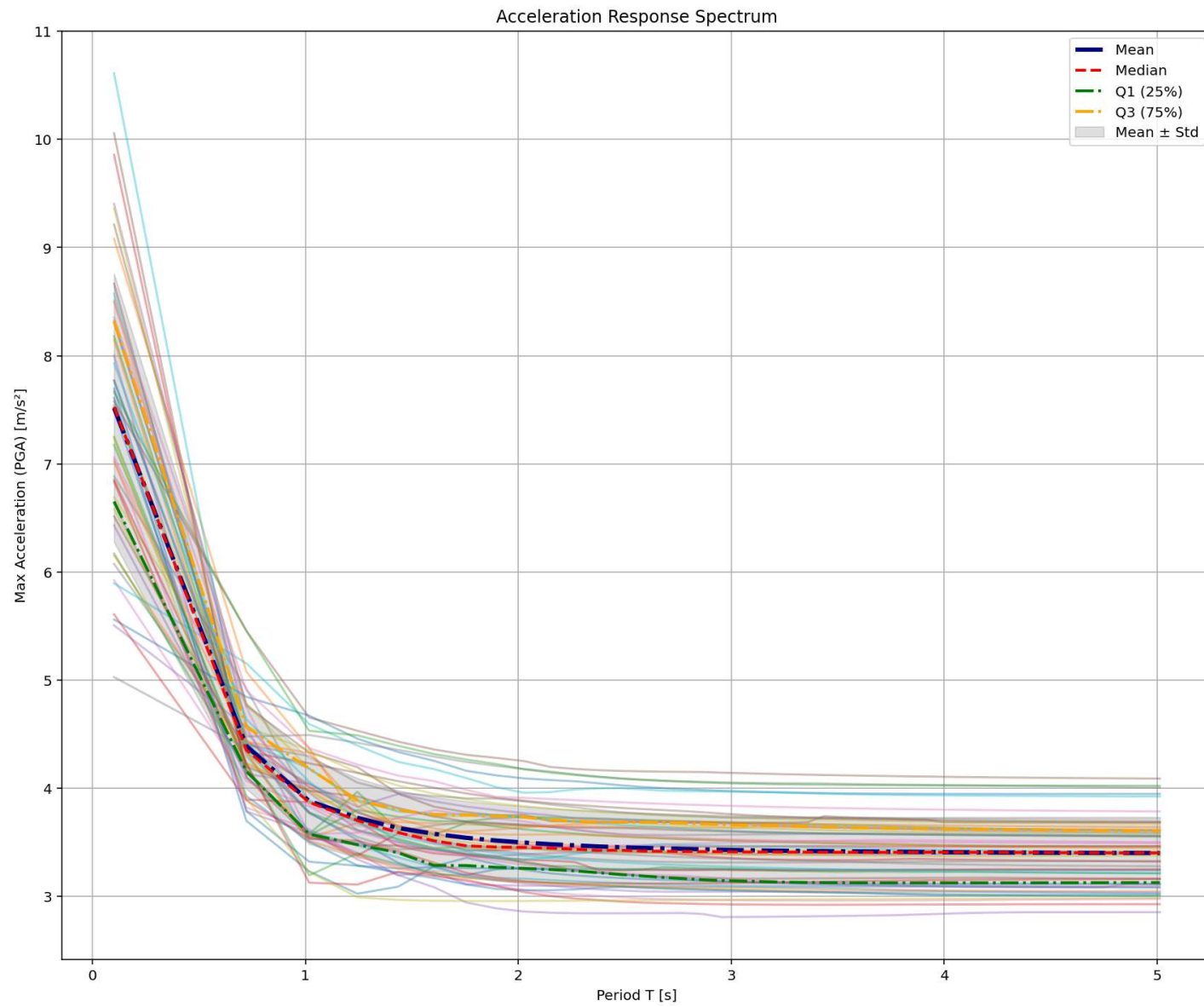
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # NONLINEAR DYNAMIC ANALYSIS UNDER MULTI GROUND MOTION RECORDS COMPUTATION AND VISUALIZATI
4 # RESPONSE SPECTRA OF ACCELERATION, VELOCITY, DISPLACEMENT DUCTILITY DAMAGE INDEX USING OP
5 #
6 # CONSTANT STRUCTURAL DUCTILITY RATIO RESPONSE SPECTRUM
7 #
8 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
9 #
10 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
11 # EMAIL: salar.d.ghashghaei@gmail.com
12 #####
13
14 """
15 This code implements a comprehensive nonlinear dynamic analysis framework for
16 performance-based earthquake engineering assessment of single-degree-of-freedom
17 (SDOF) systems. The methodology combines traditional nonlinear time-history
18 analysis with modern probabilistic and machine learning techniques for advanced
19 structural performance evaluation with changing structural ductility ratio and over strength fa
20
21 KEY ENGINEERING OBJECTIVES:
22 1. Comparative assessment of hysteretic models for seismic response prediction
23 2. Probabilistic seismic demand analysis using multiple ground motions
24 3. Development of fragility curves for performance-based earthquake engineering
25 4. Integration of data science methods for structural reliability assessment
26
27 ANALYTICAL FEATURES:
28 - Nonlinear material behavior with pinching and degradation
29 - Response spectrum analysis across period range
30 - Real-time structural health monitoring metrics
31 - Statistical characterization of seismic demands
32 - Machine learning-based damage prediction
33
34 """

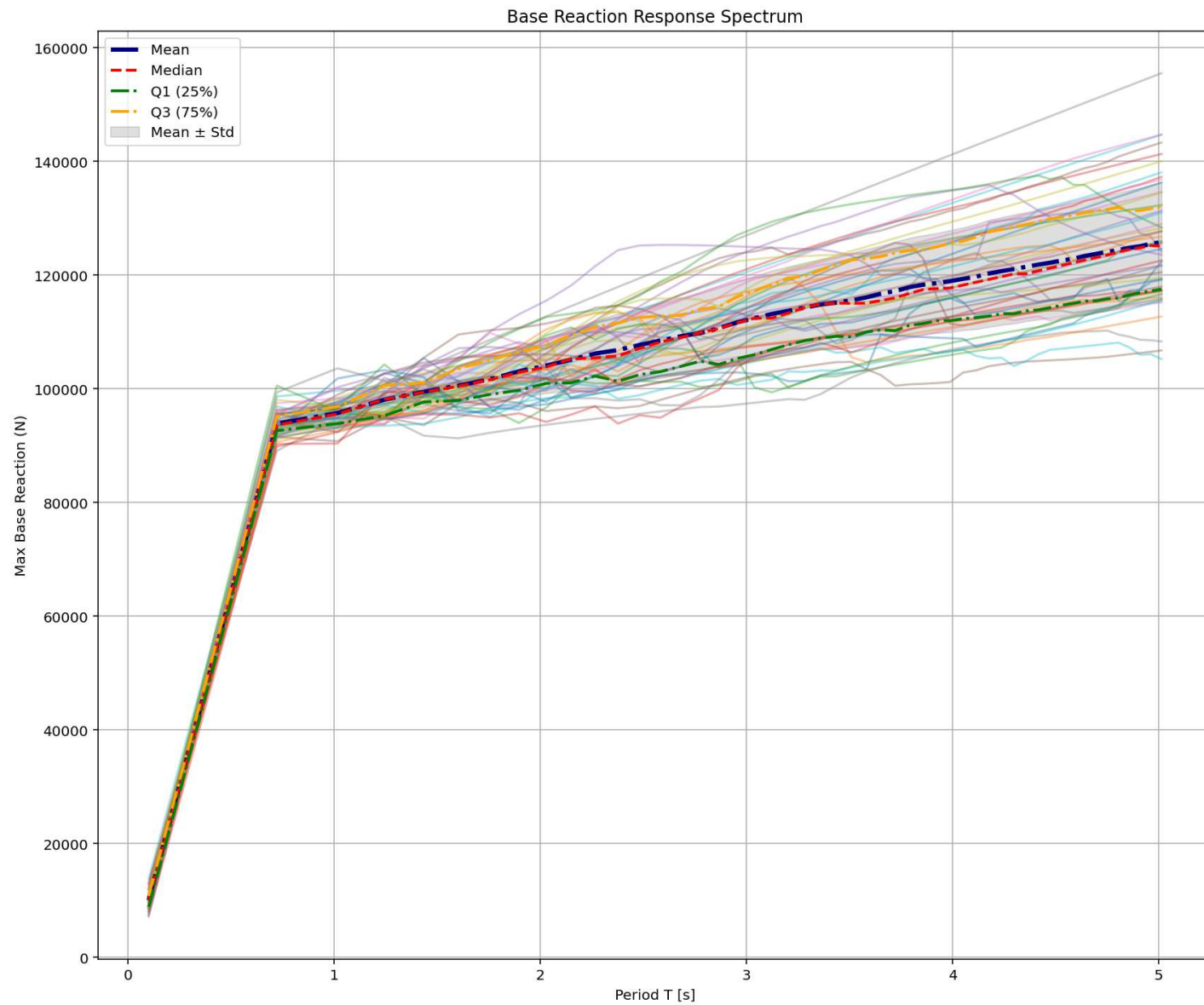
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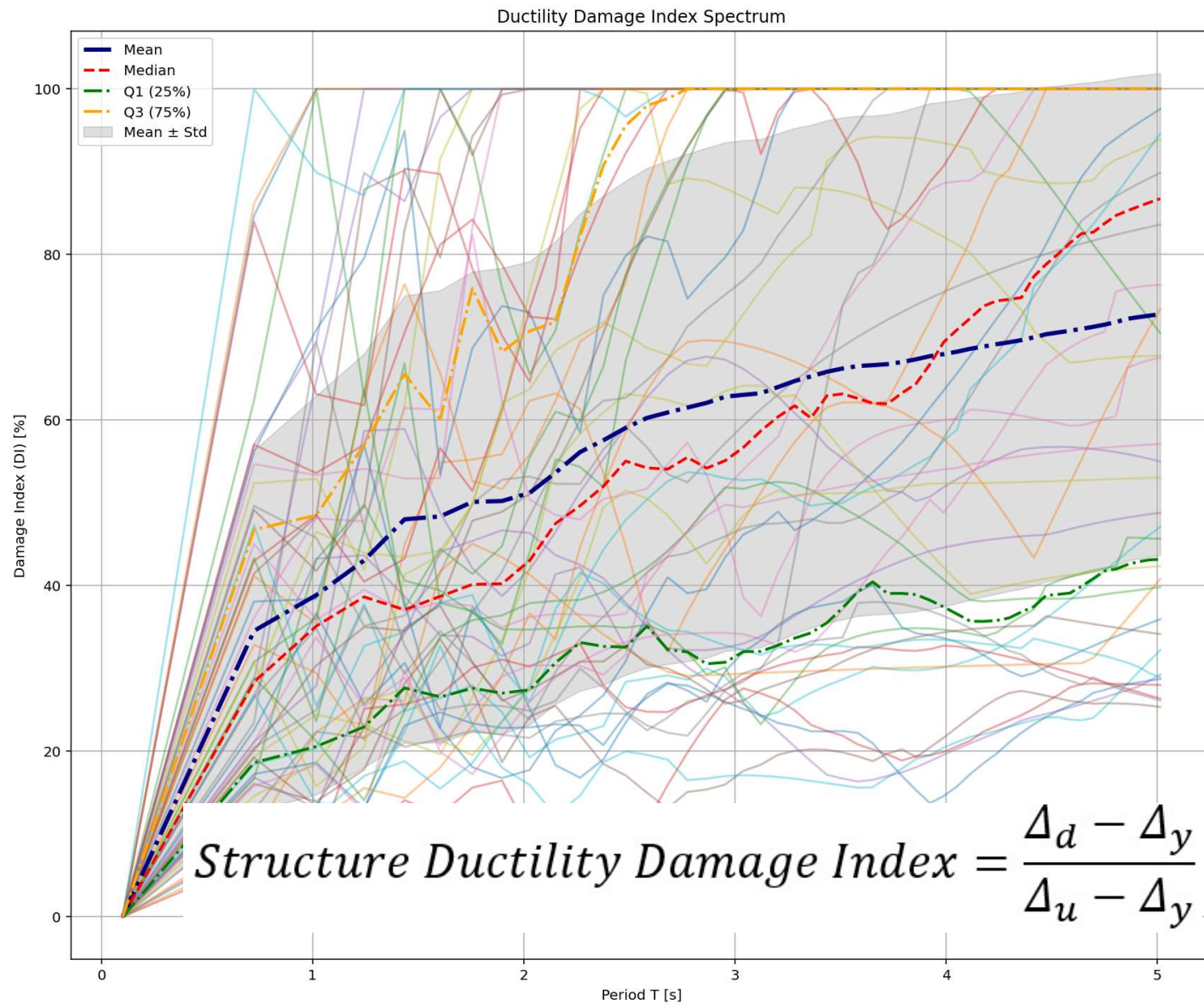


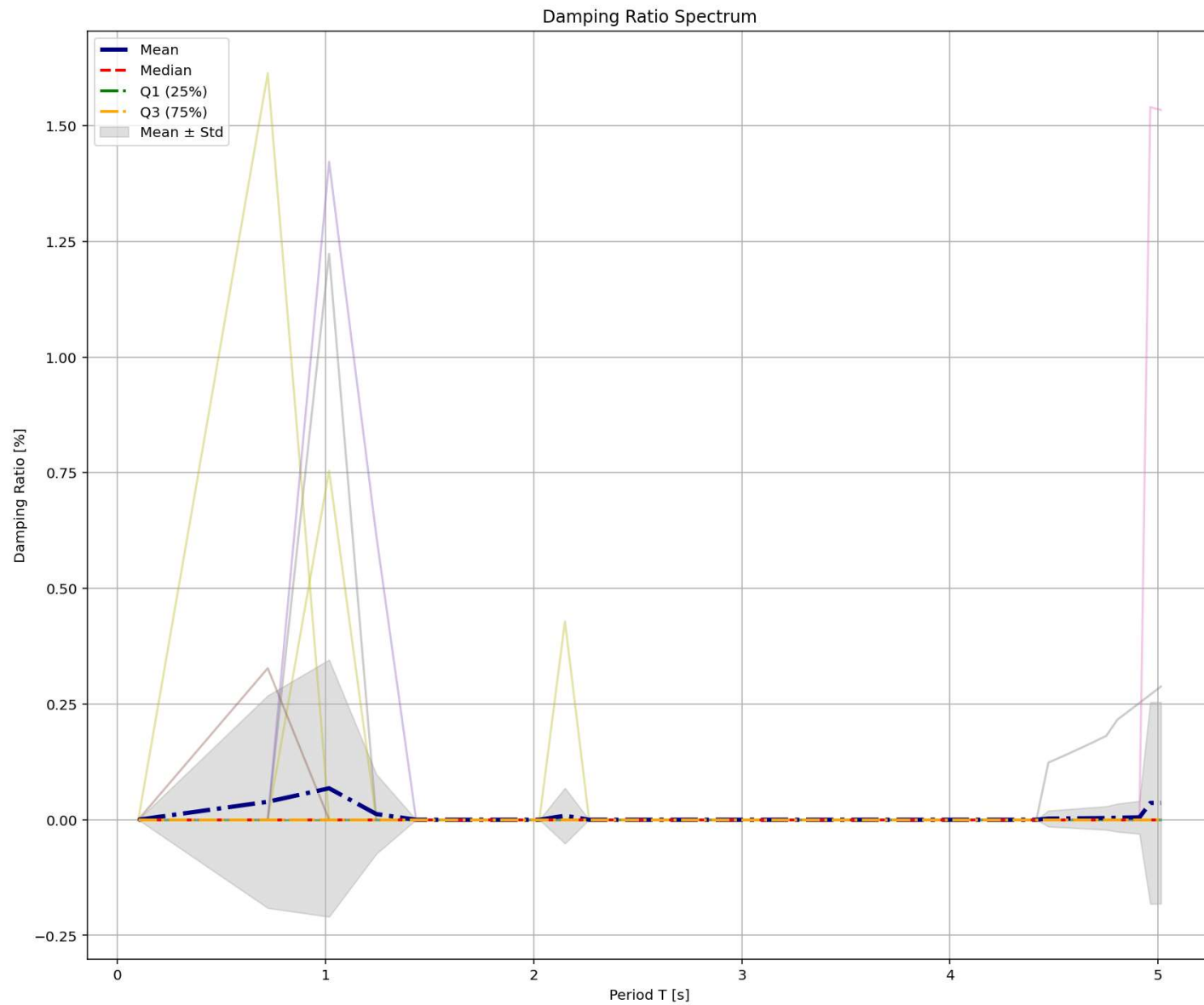


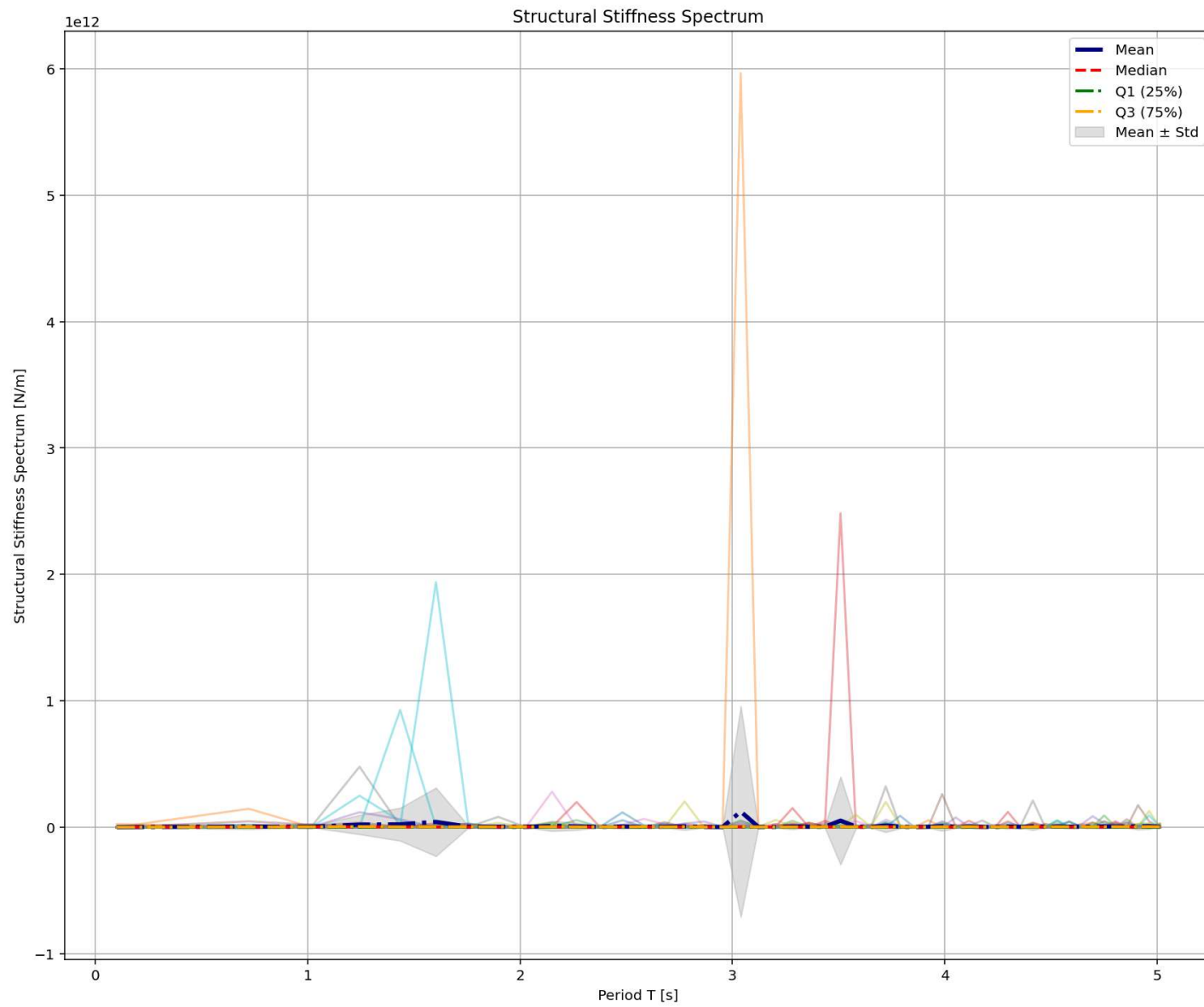




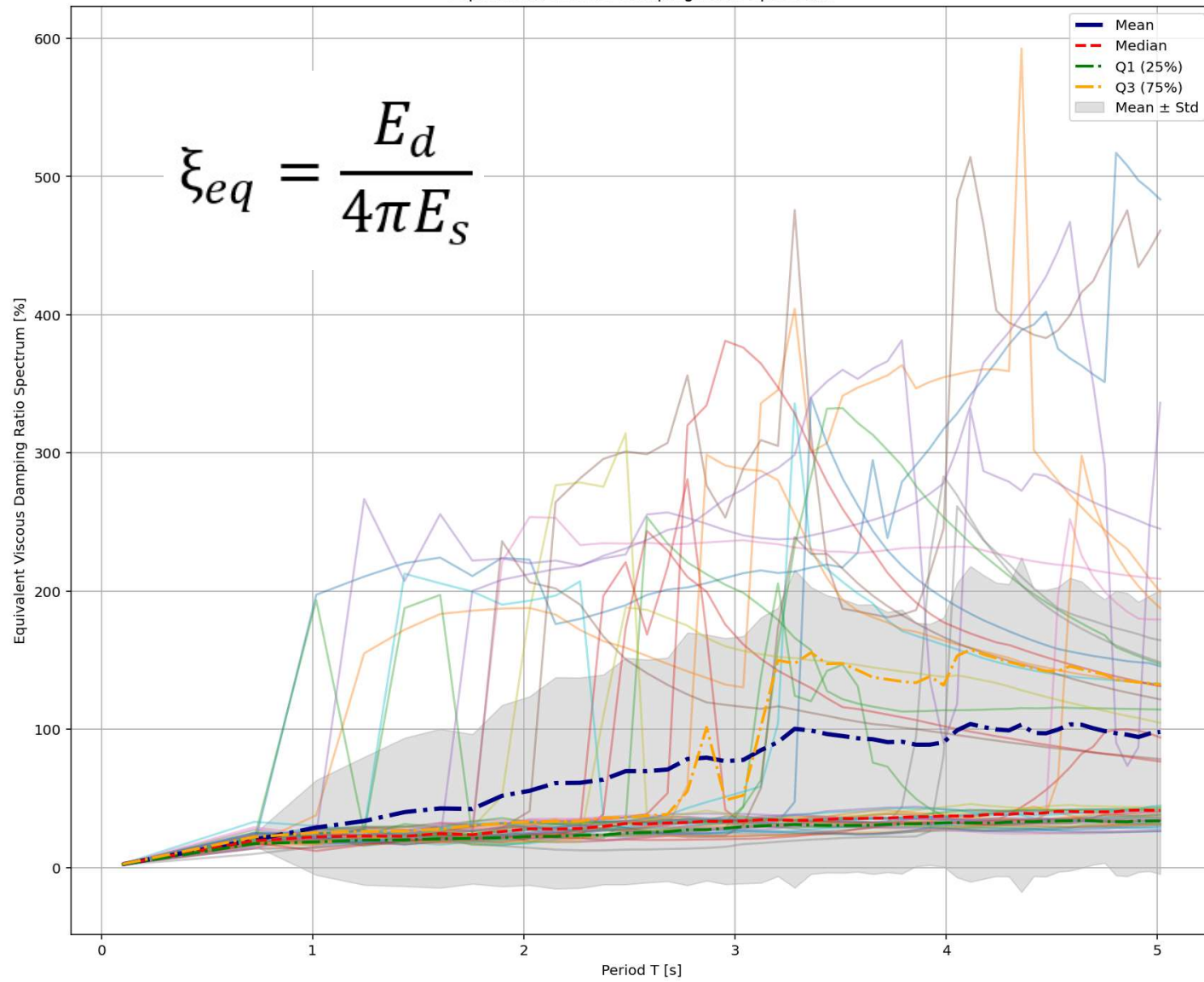




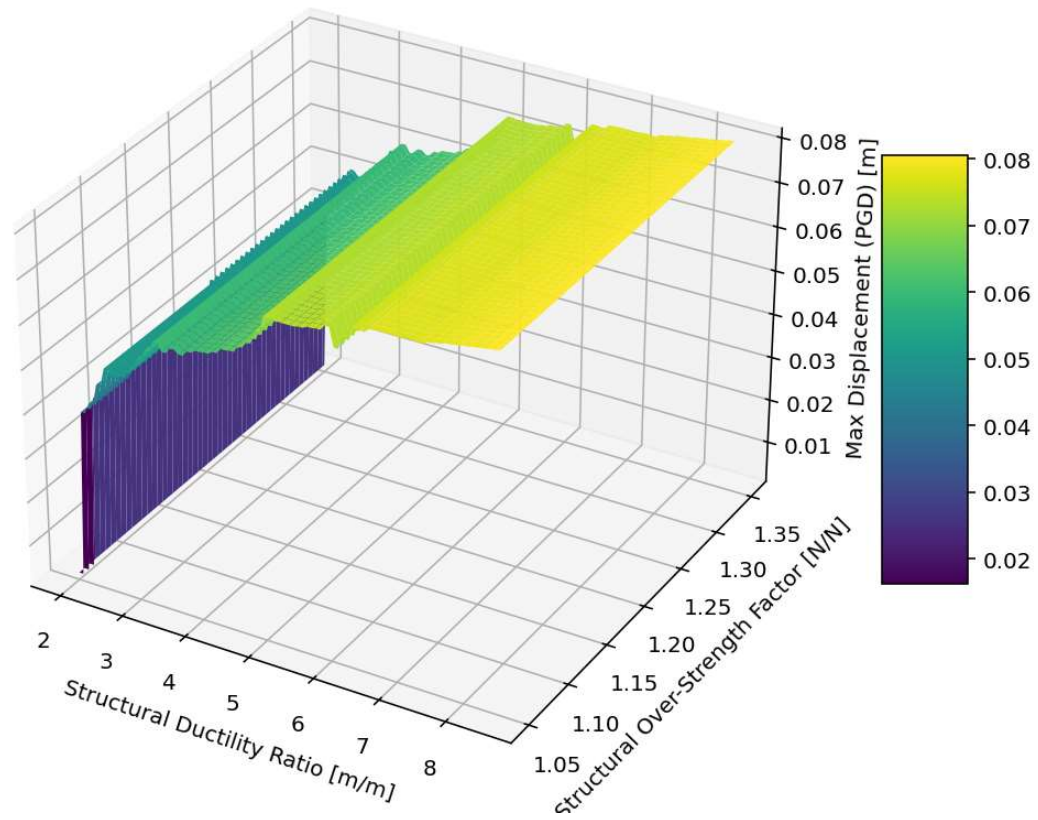




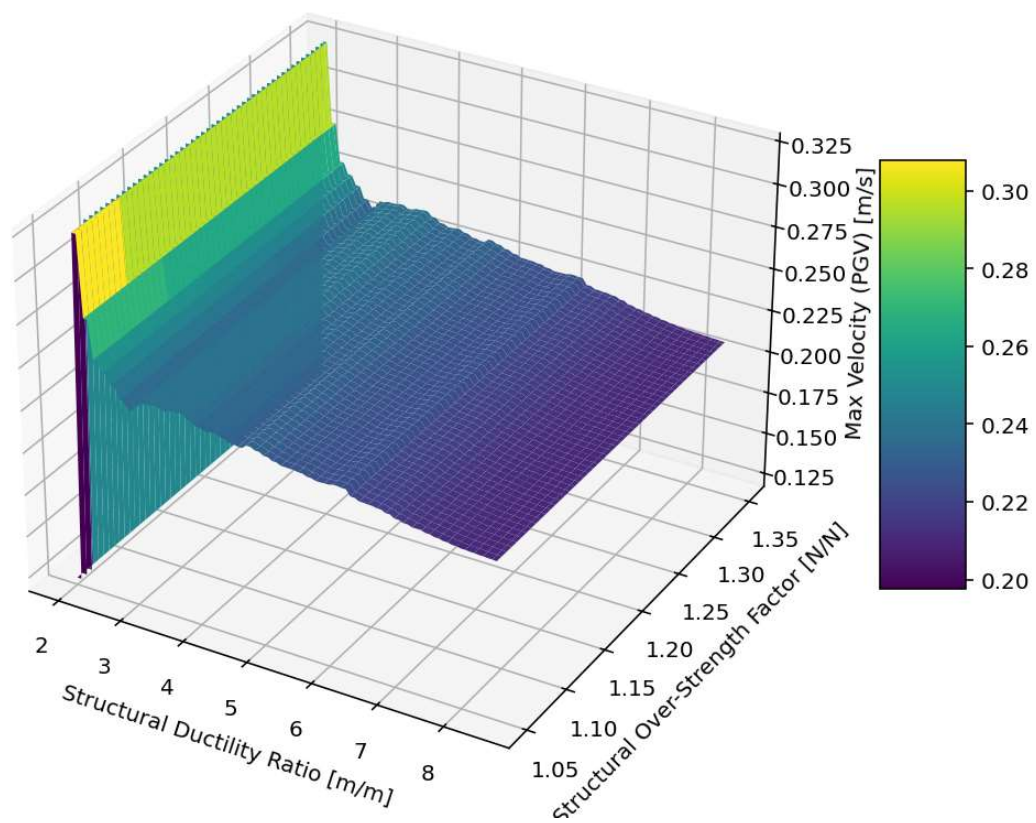
Equivalent Viscous Damping Ratio Spectrum



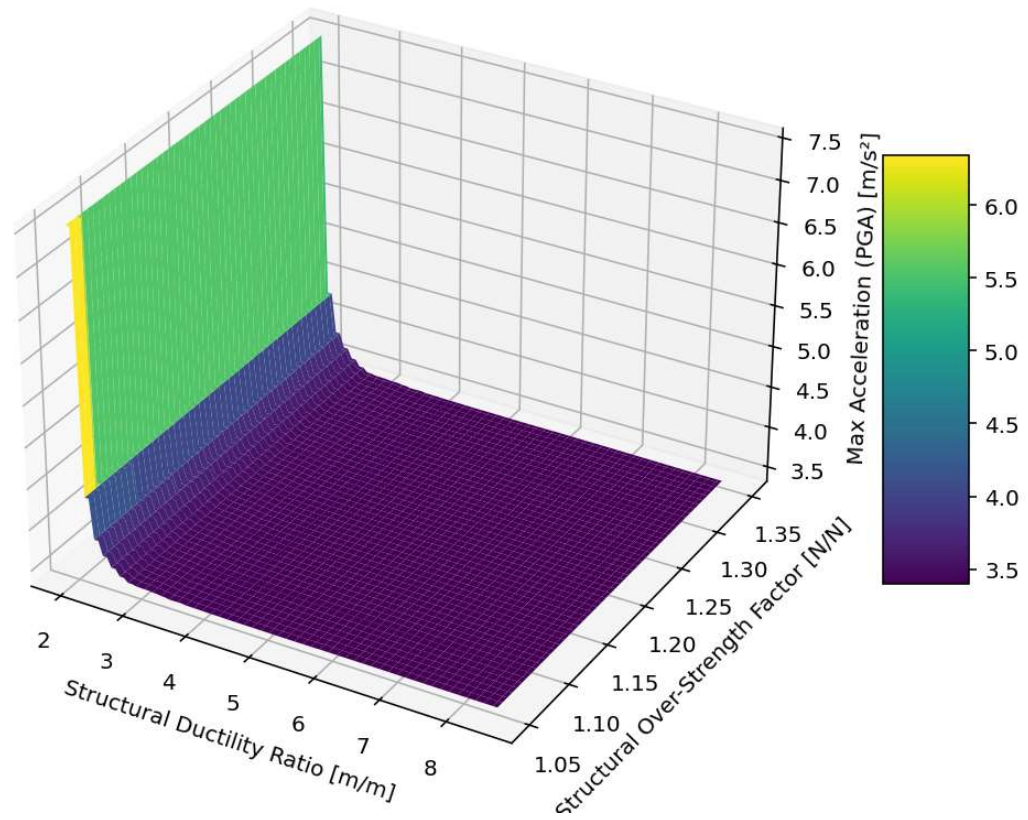
3D Contour Plot of Max Displacement (PGD) [m]



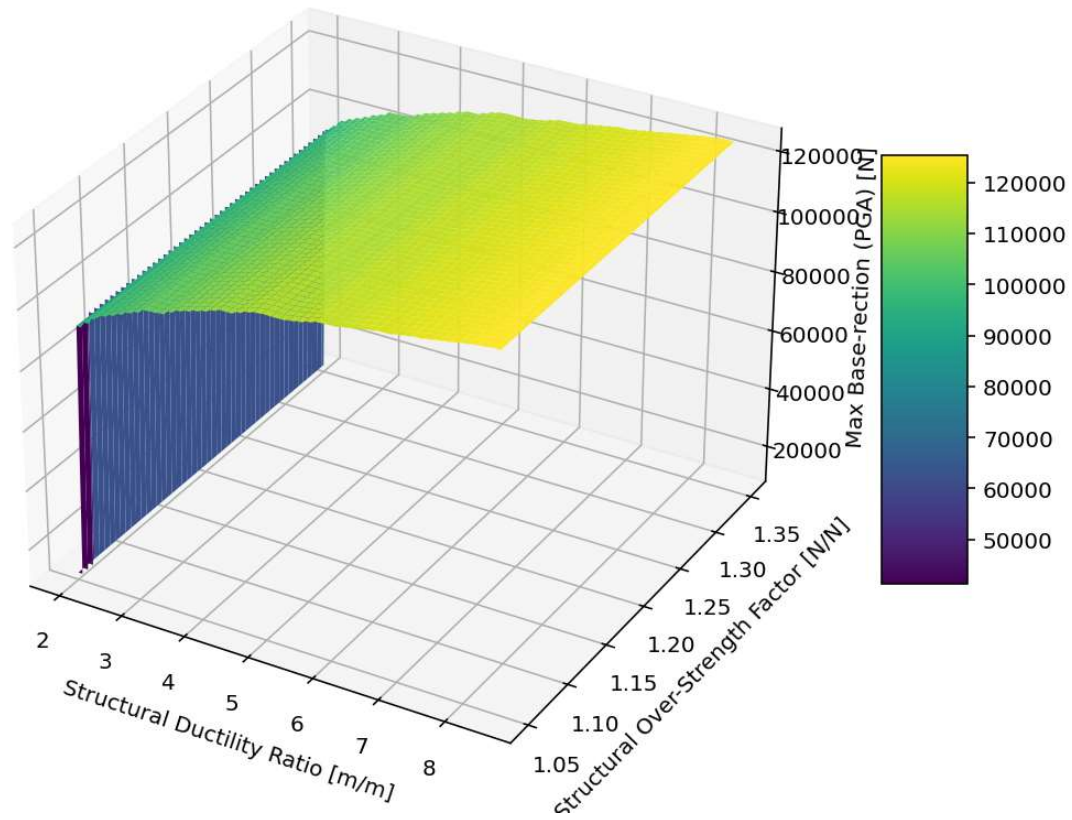
3D Contour Plot of Max Velocity (PGV) [m/s]



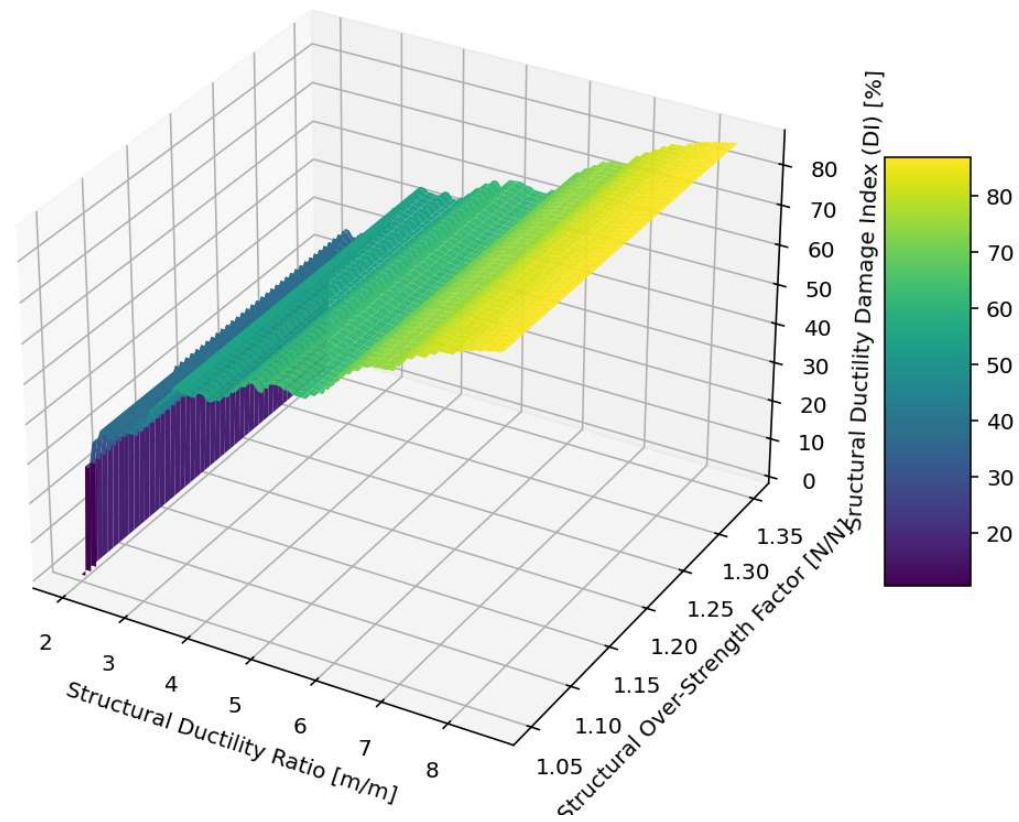
3D Contour Plot of Max Acceleration (PGA) [m/s^2]



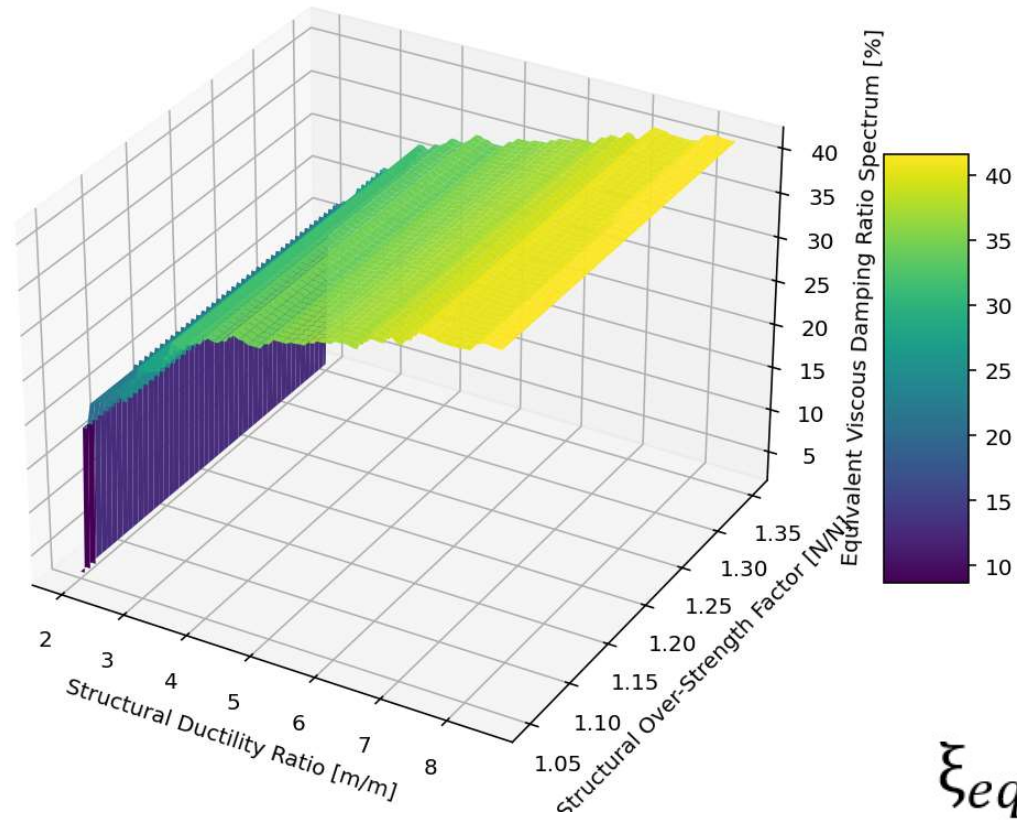
3D Contour Plot of Max Base-rection (PGA) [N]



3D Contour Plot of Structural Ductility Damage Index (DI) [%]

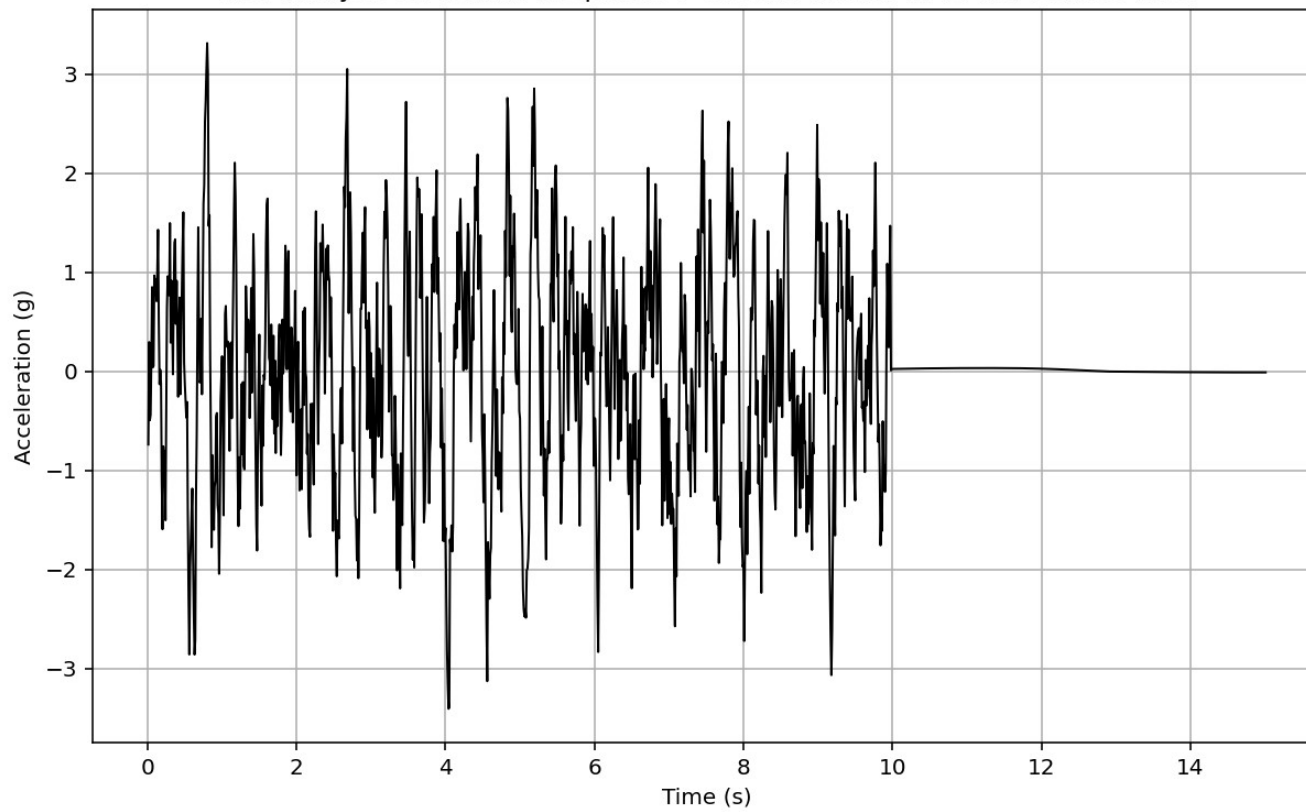


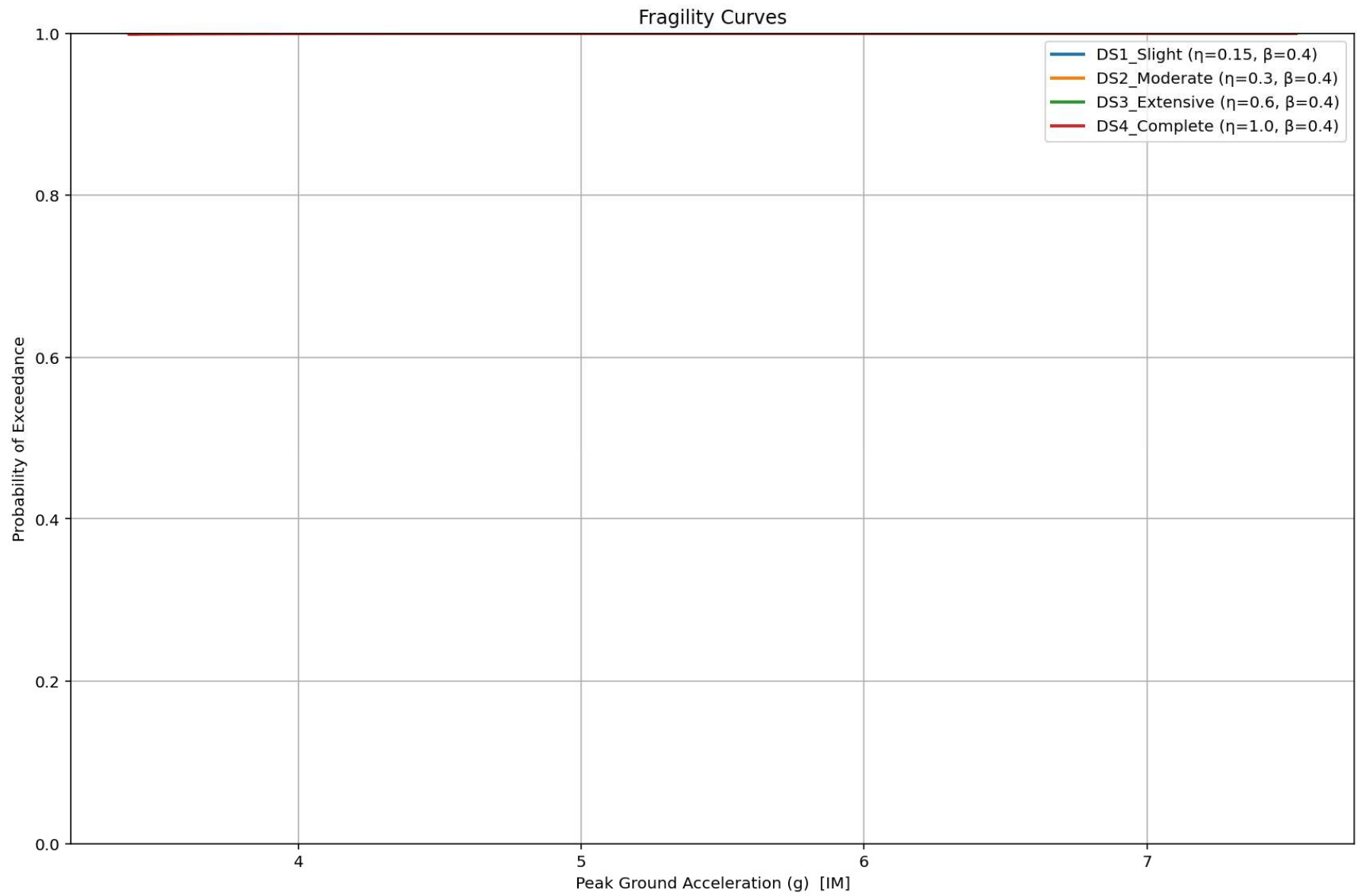
3D Contour Plot of Equivalent Viscous Damping Ratio Spectrum [%]

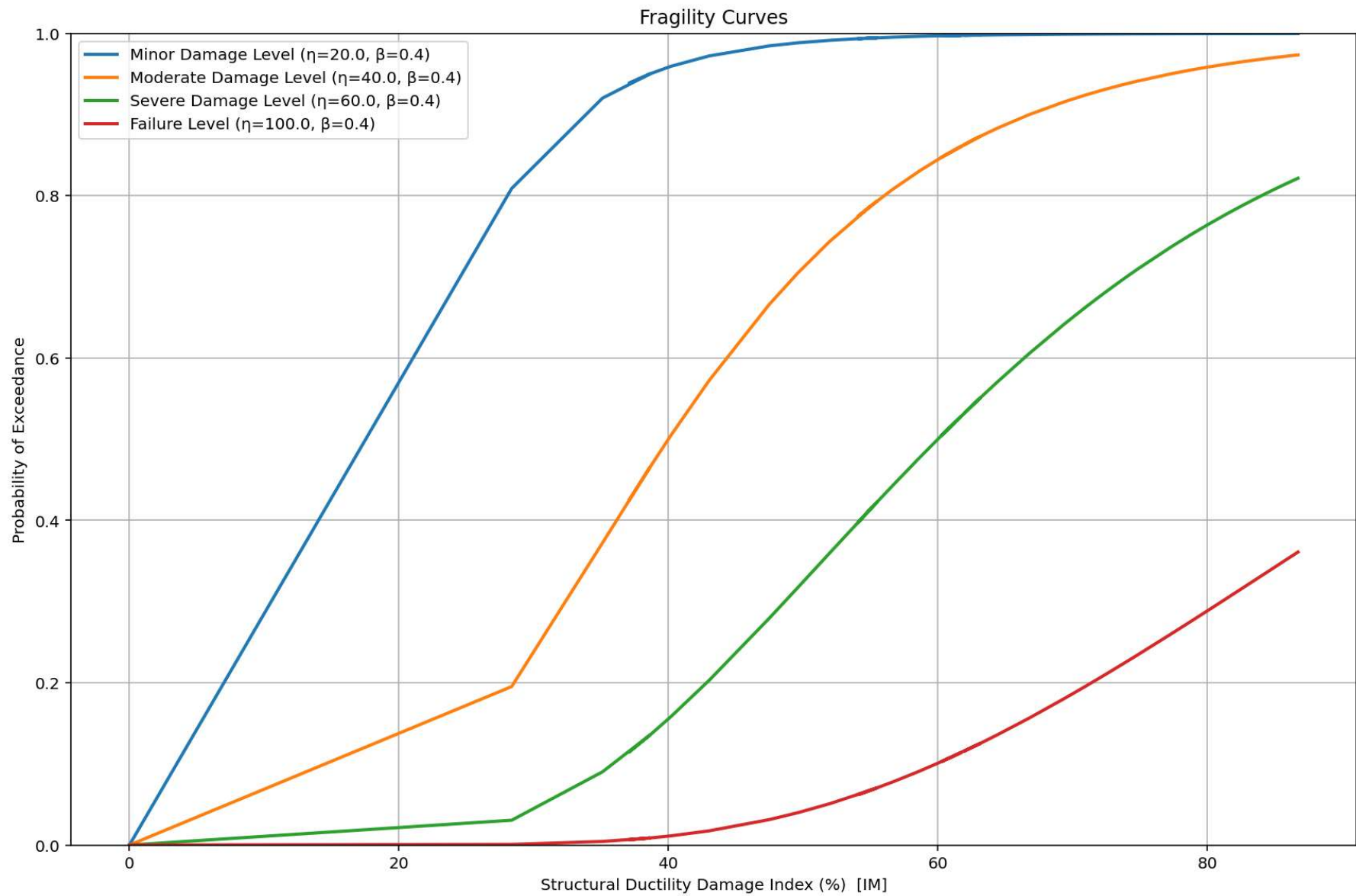


$$\xi_{eq} = \frac{E_d}{4\pi E_s}$$

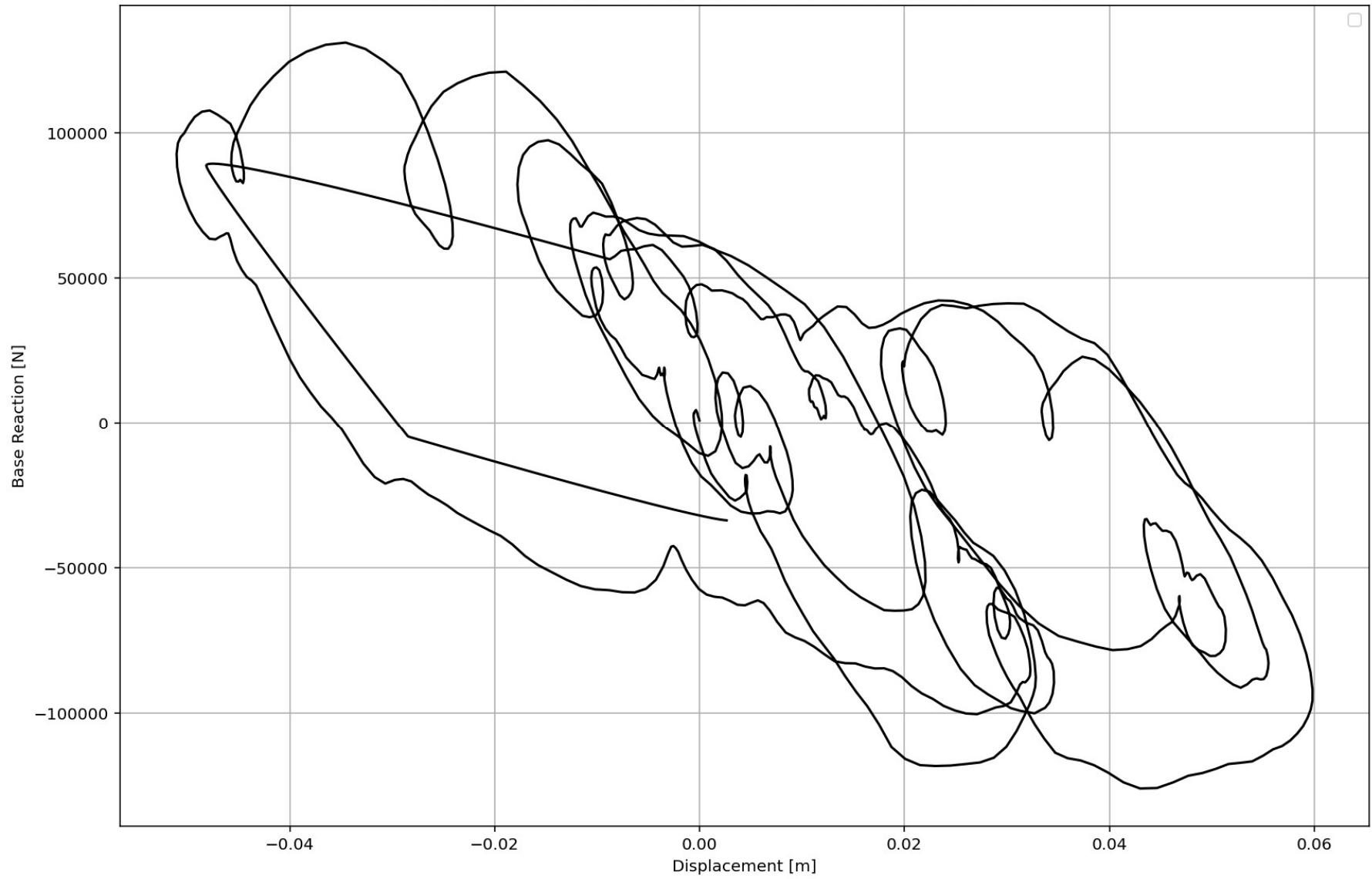
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 3.4093

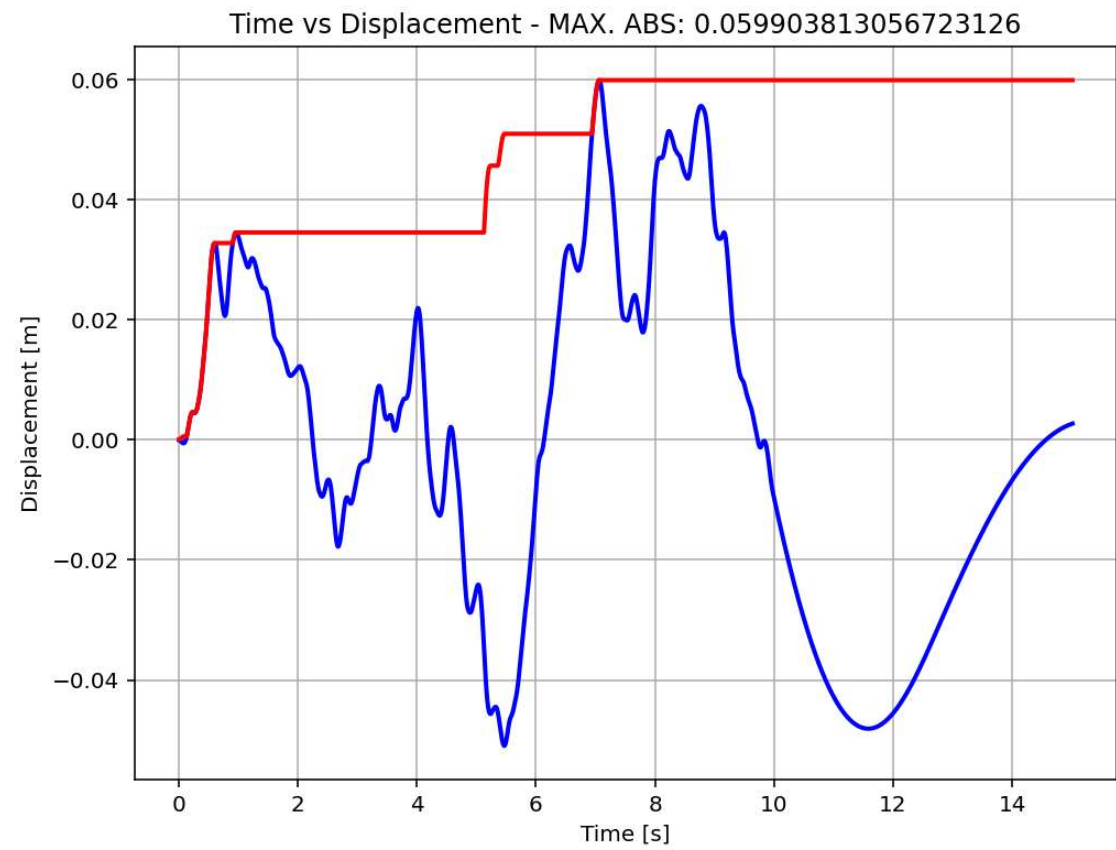


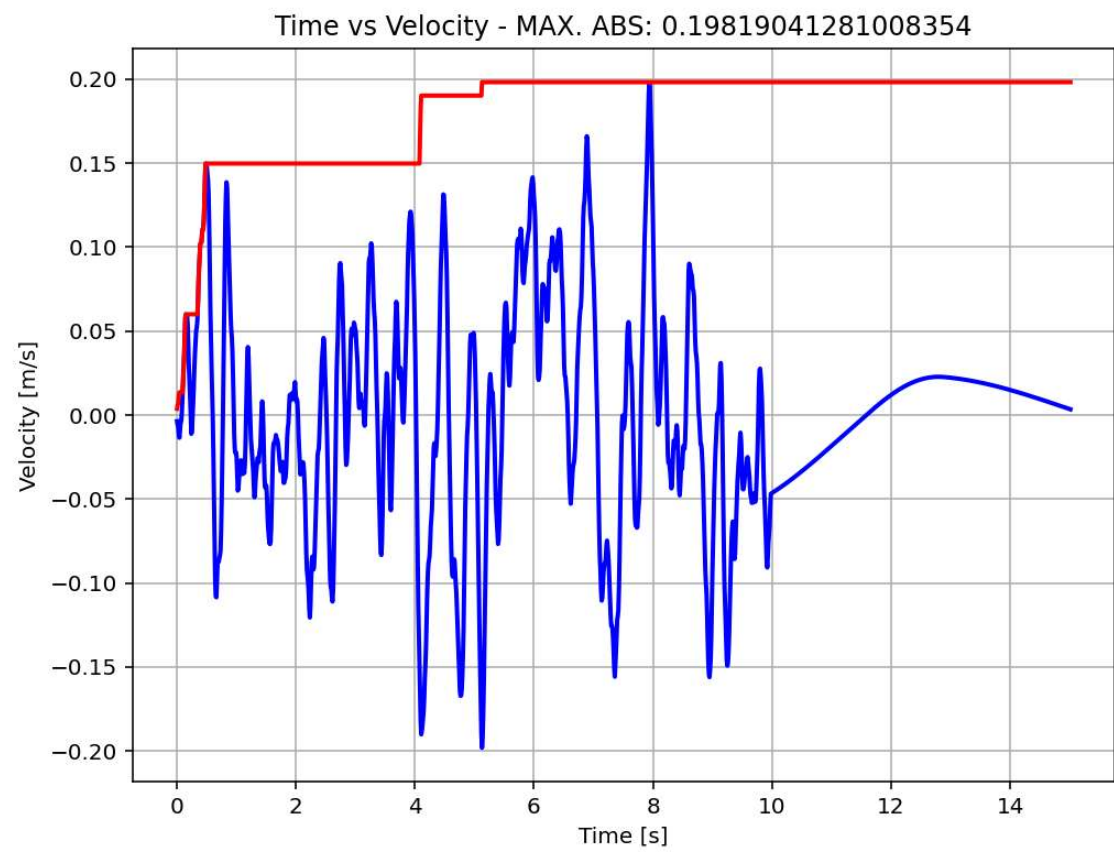




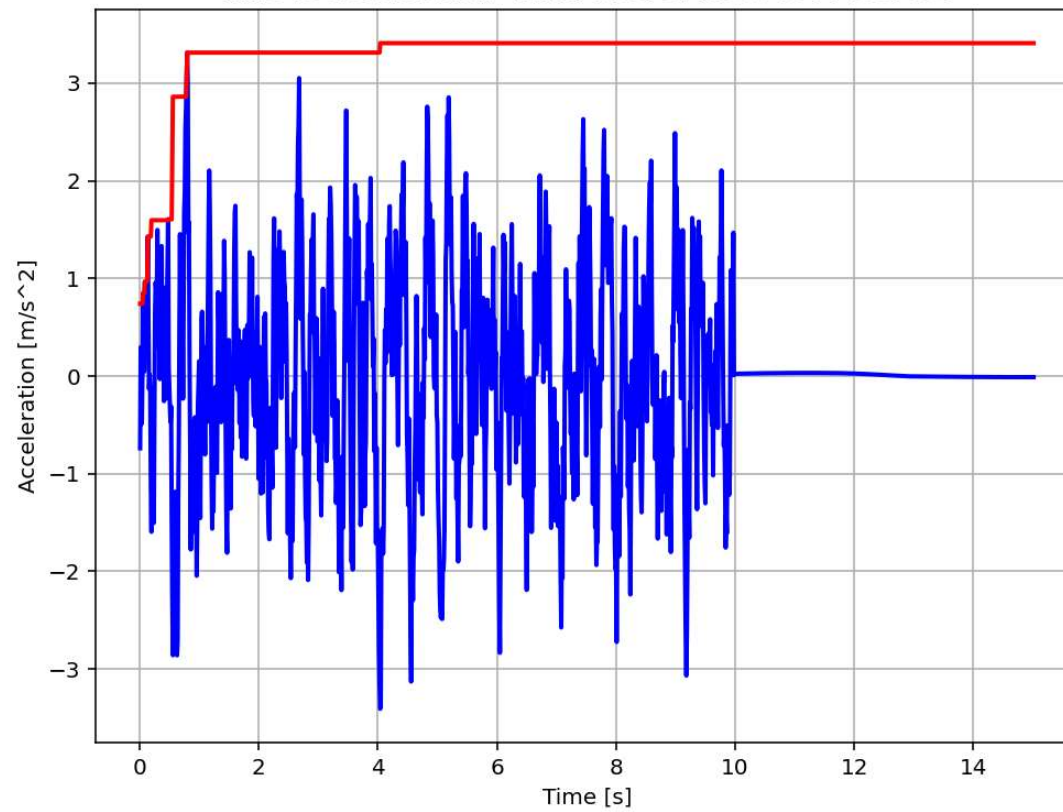
Displacement & Base Reaction Relation From Last Dynamic Analysis

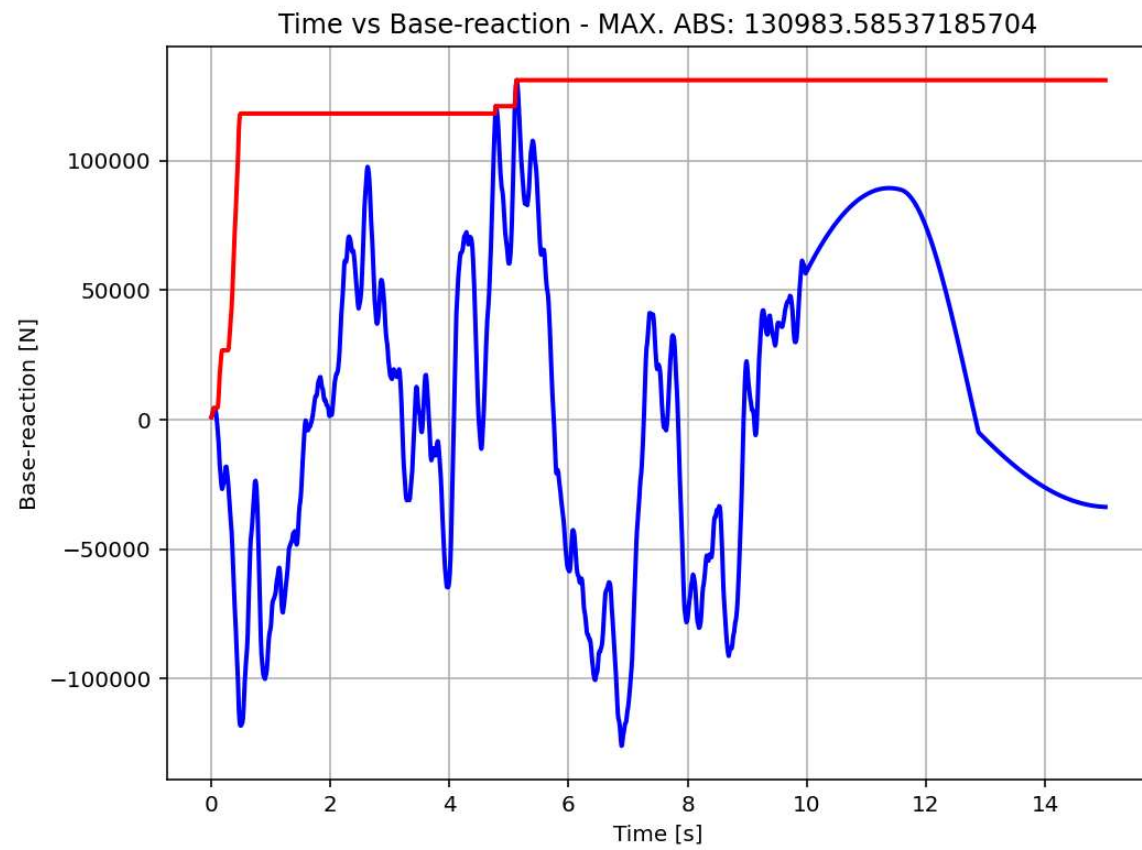


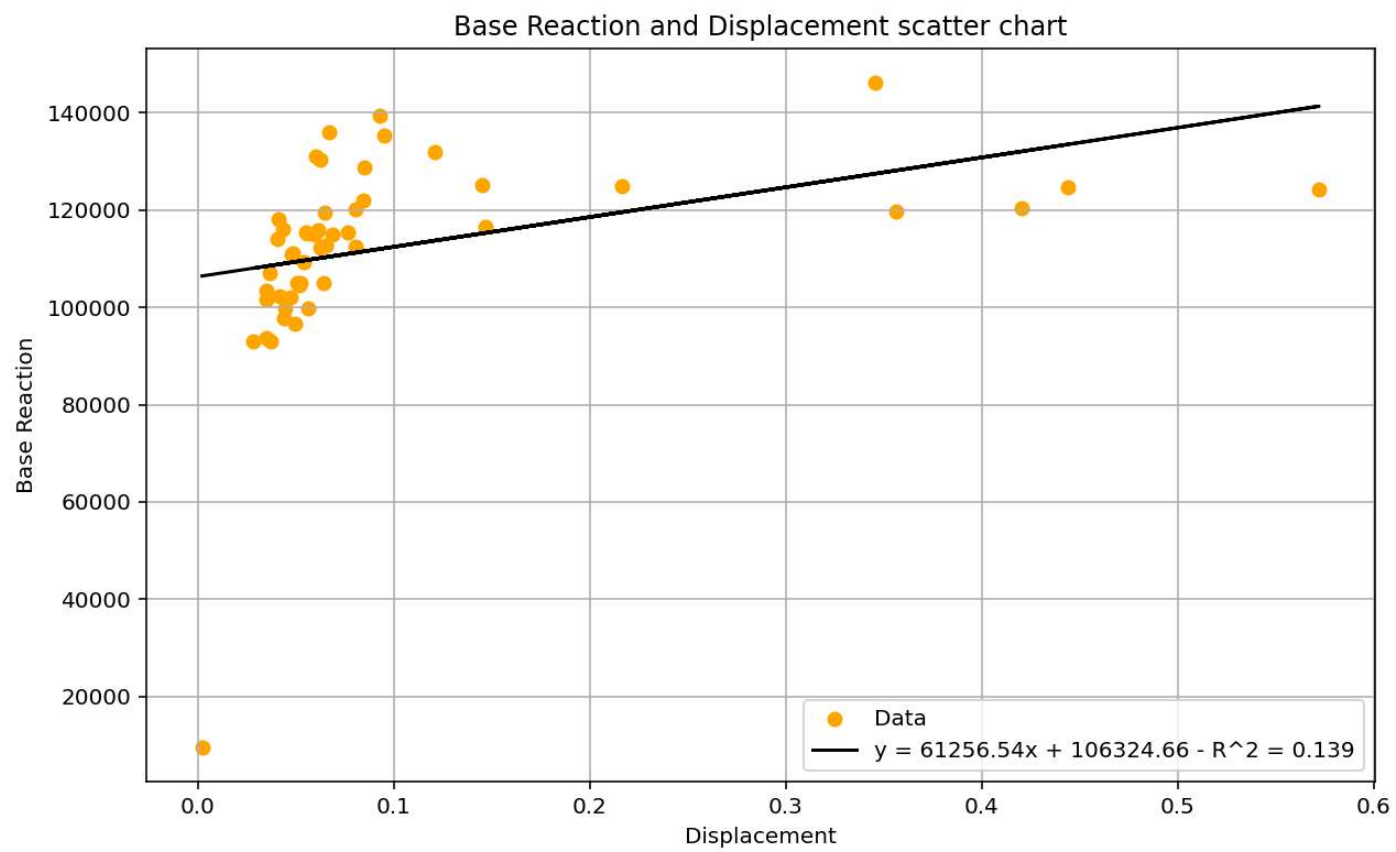


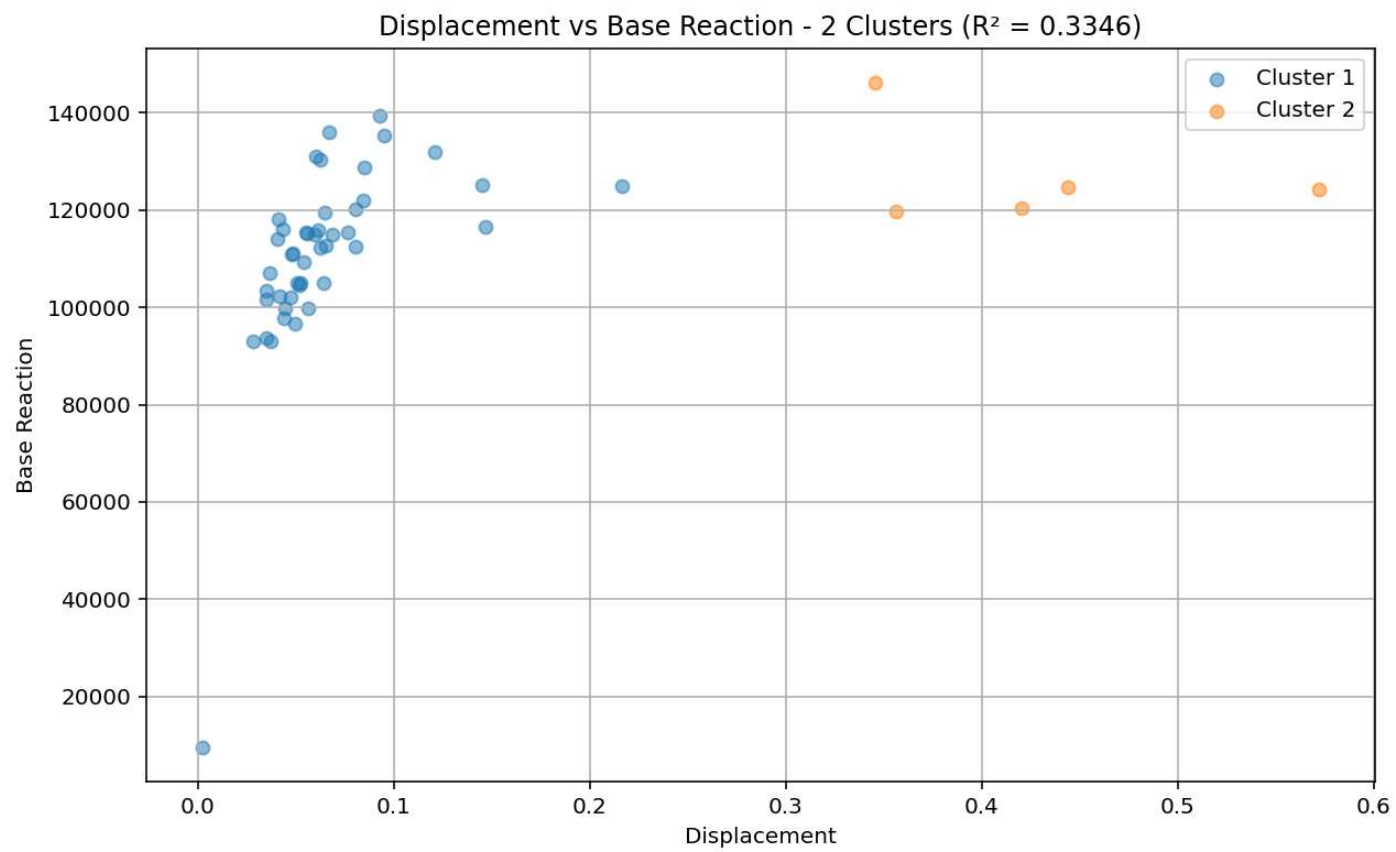


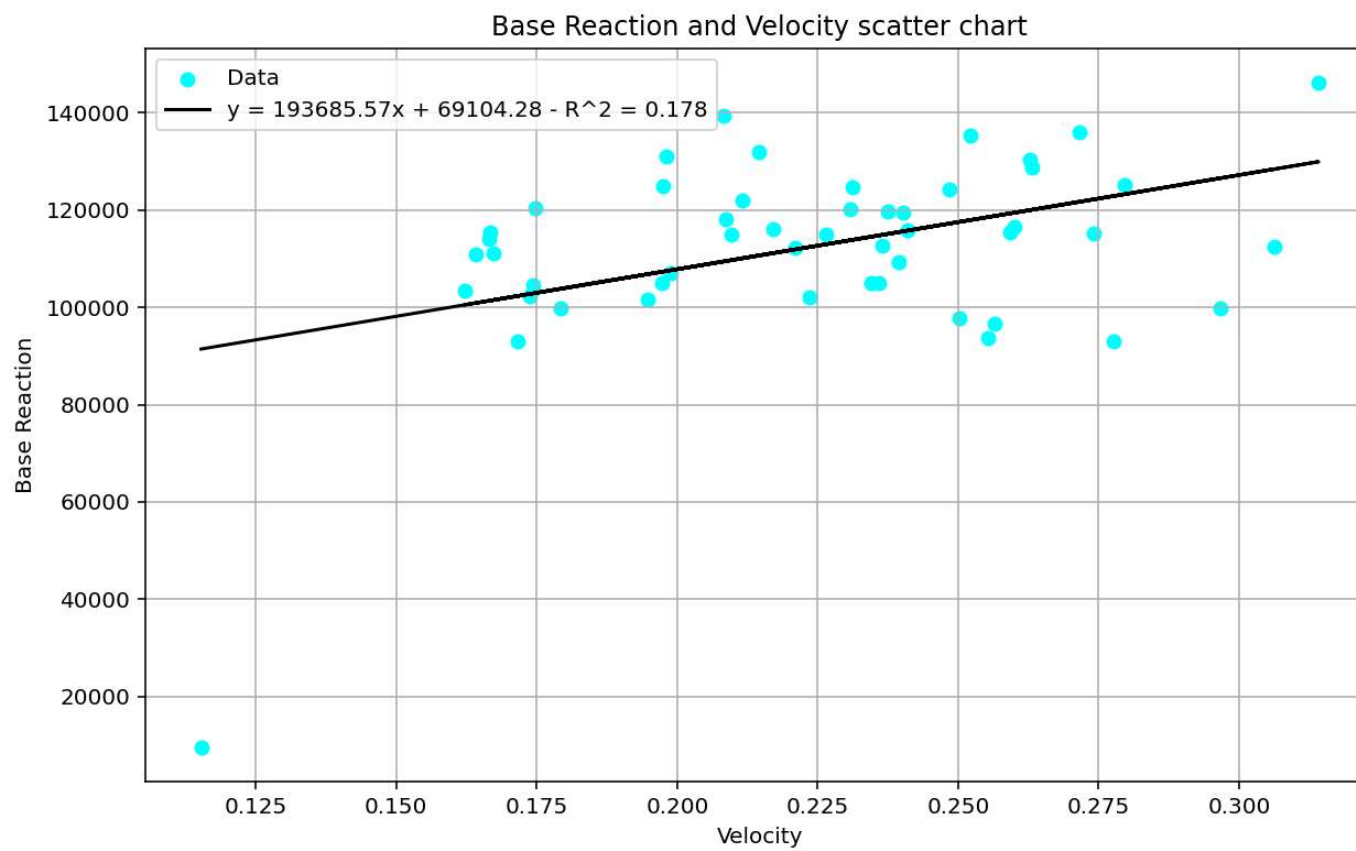
Time vs Acceleration - MAX. ABS: 3.4093343444307136

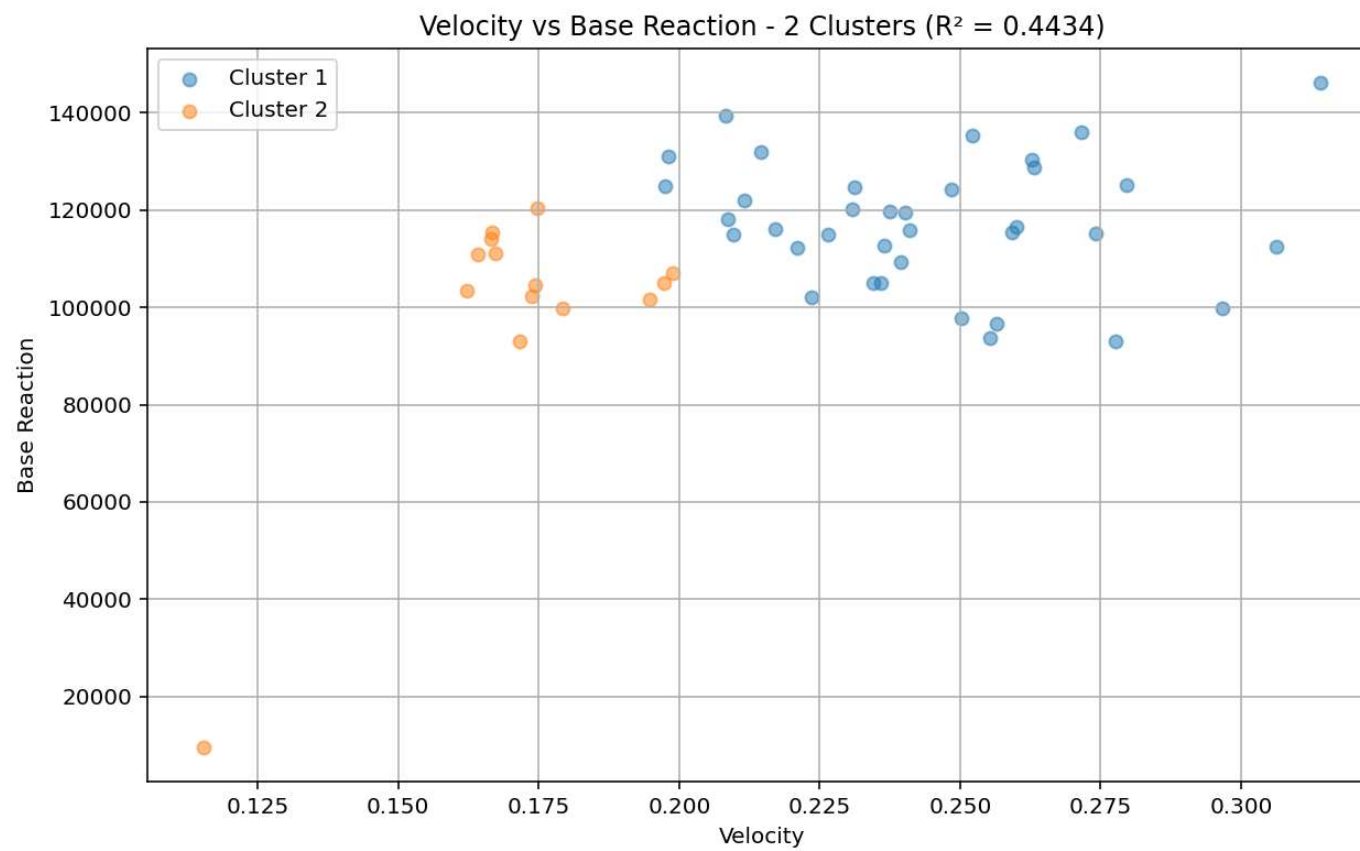


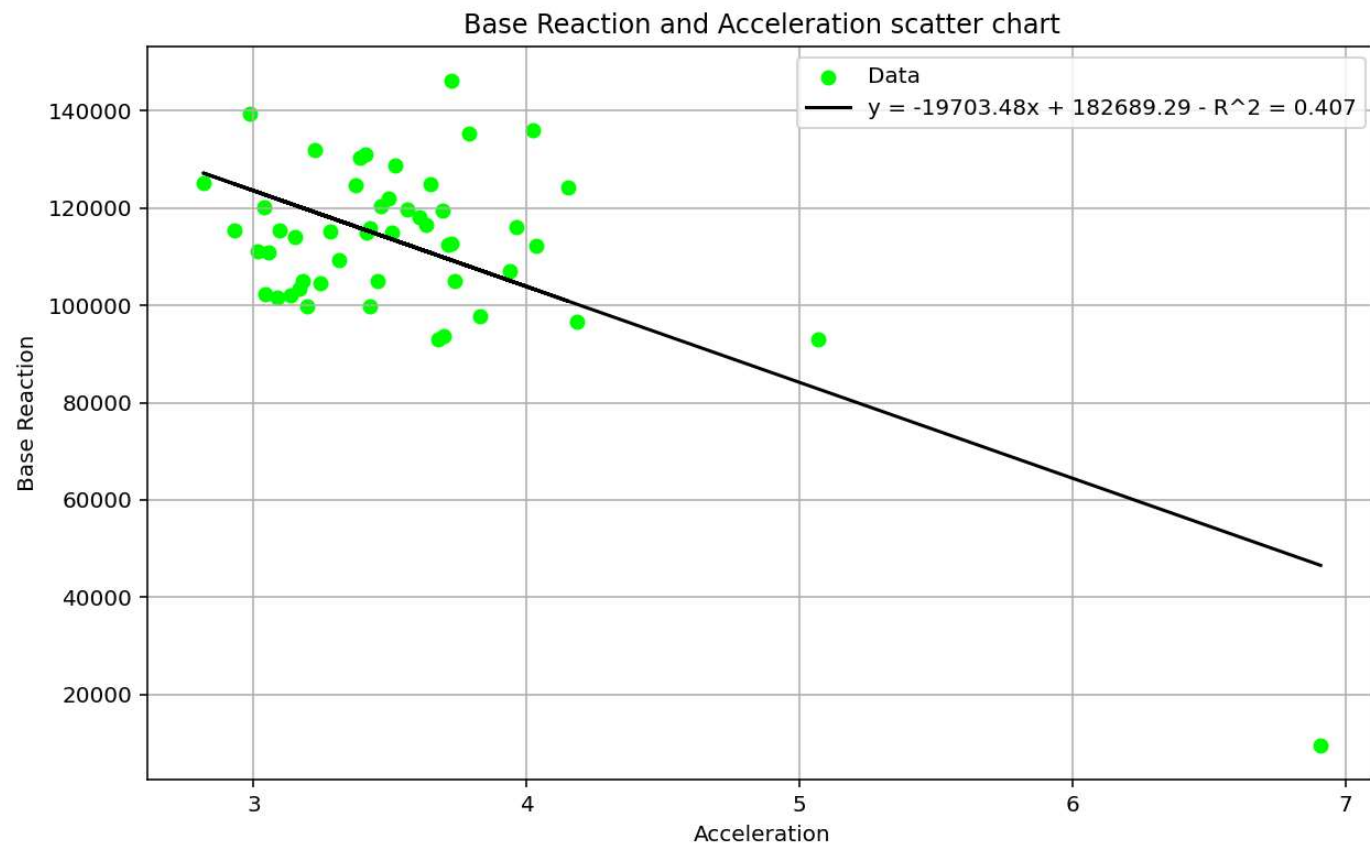


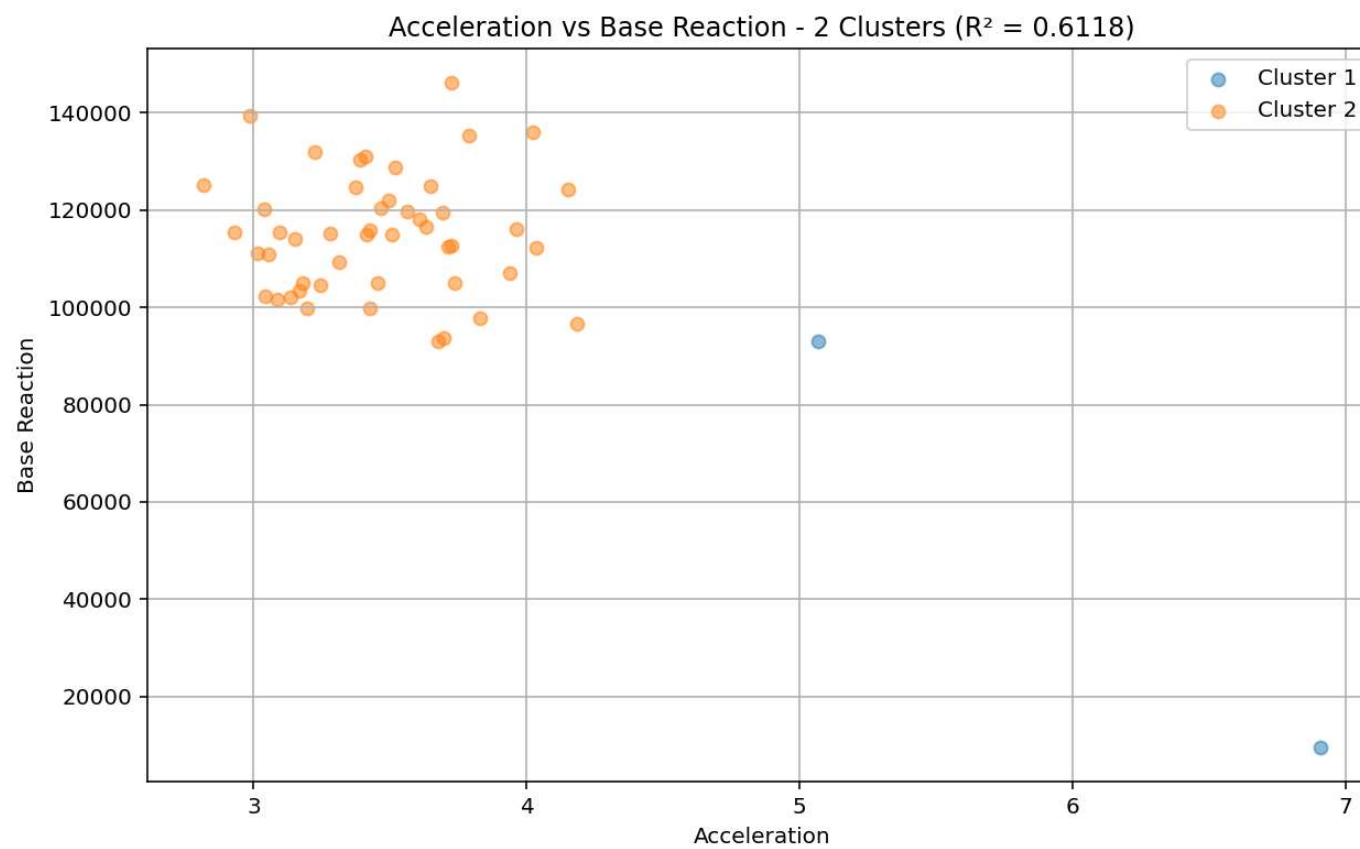


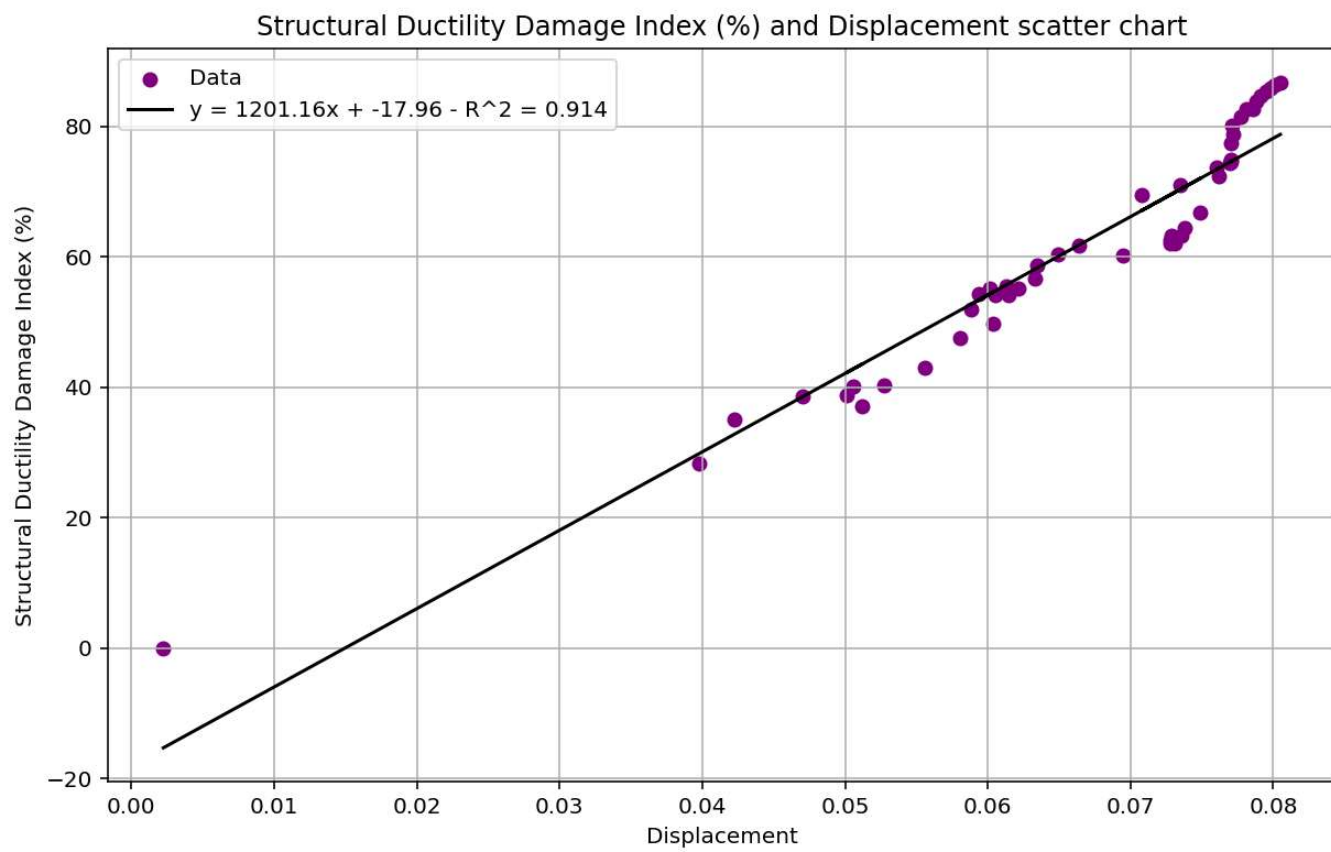


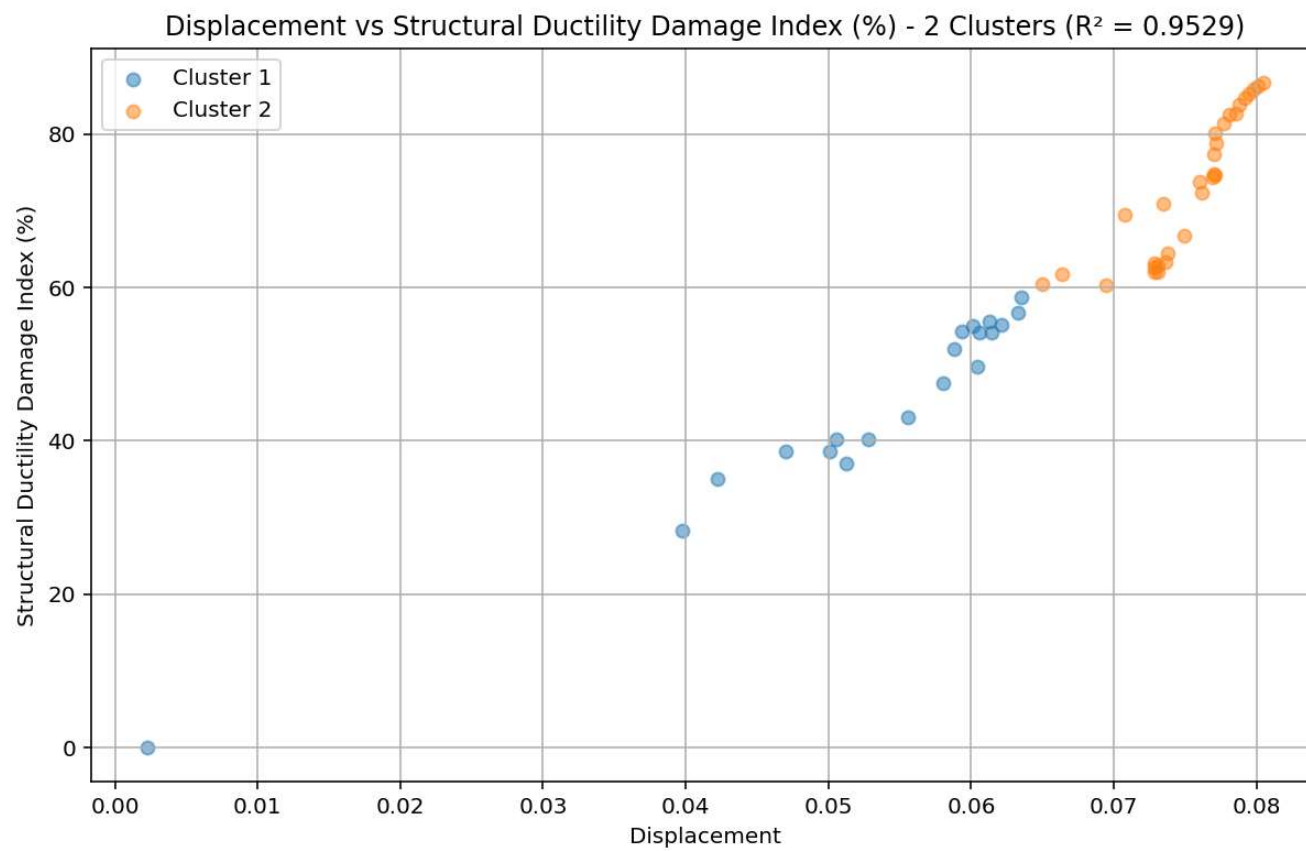


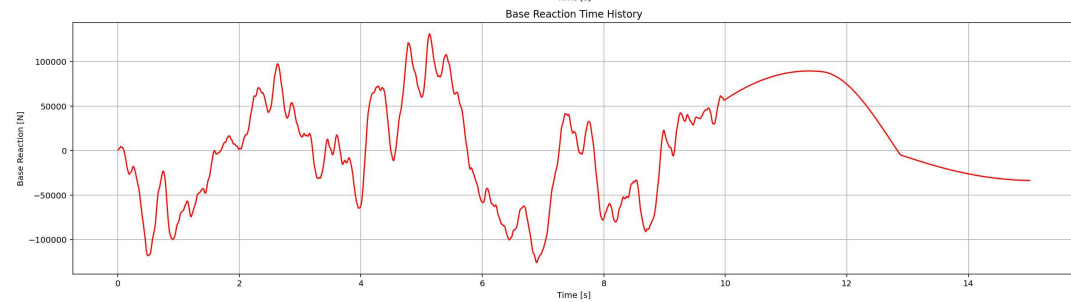
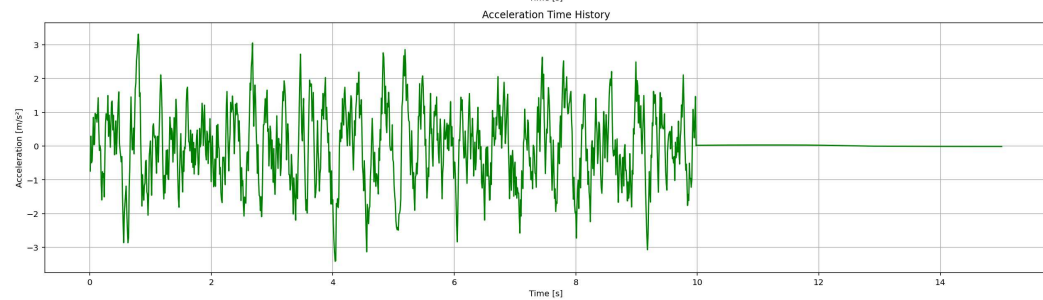
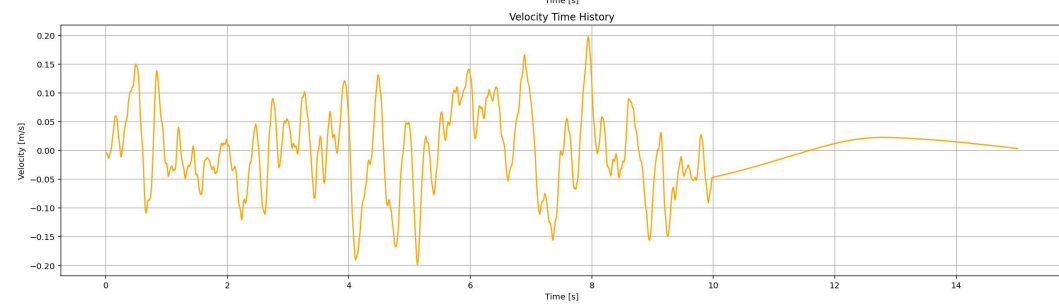
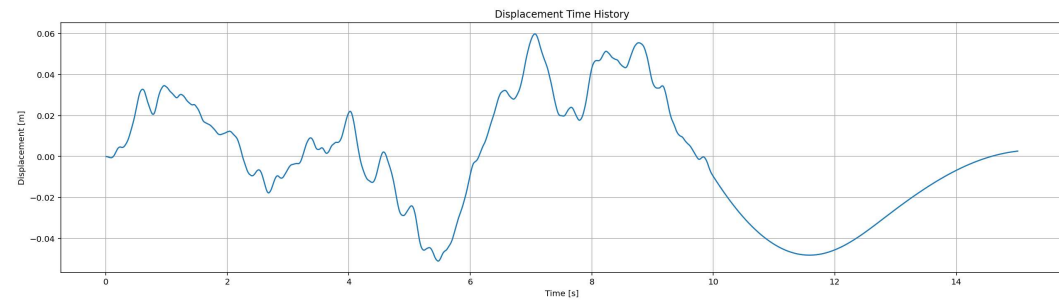


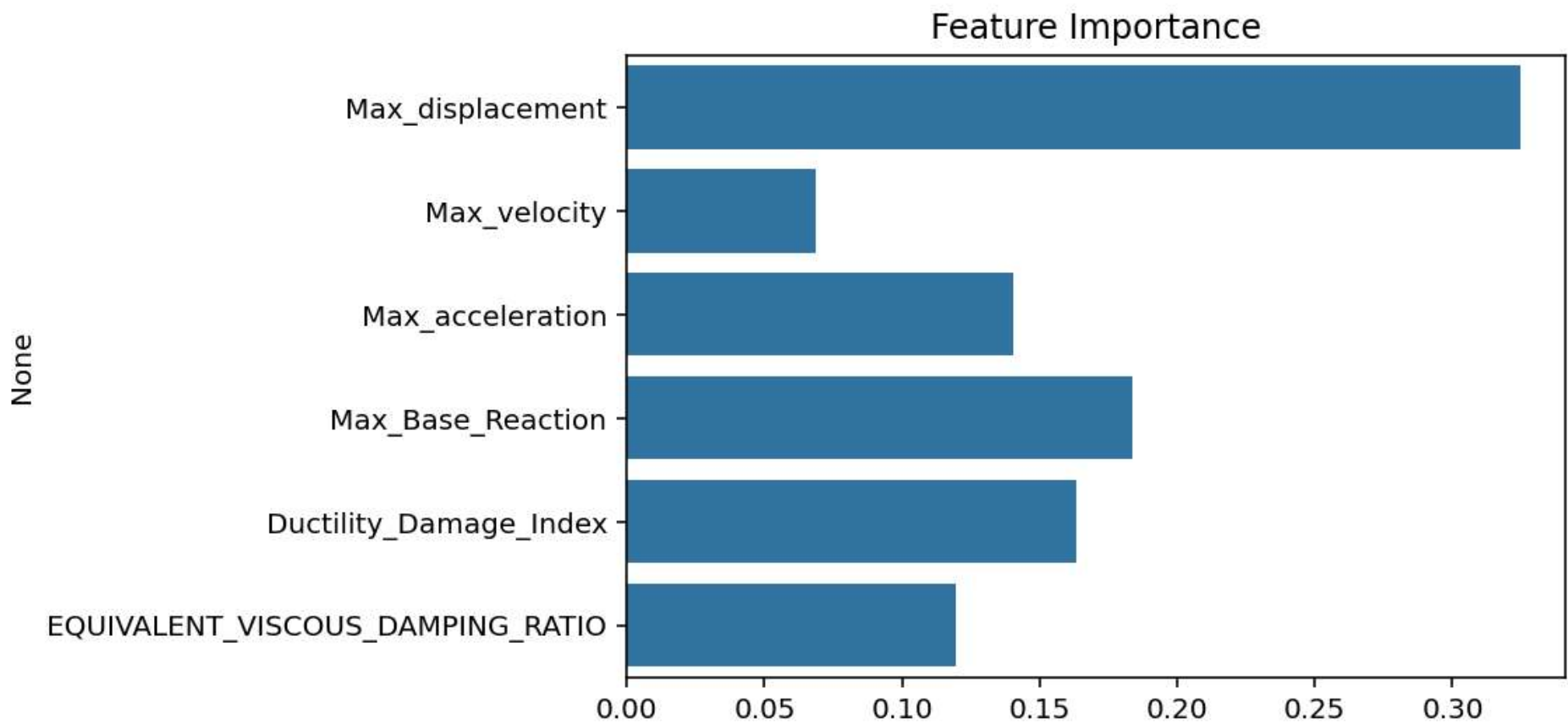












Correlation Heatmap

